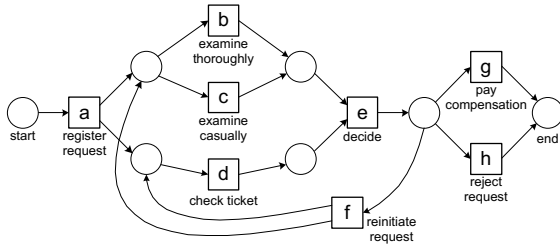


Original model is like a 5th force



initial
model

*to avoid confusion
when things change*

fitness

lift

*ability to explain
observed behavior*

precision

drag

*avoiding
underfitting*

thrust

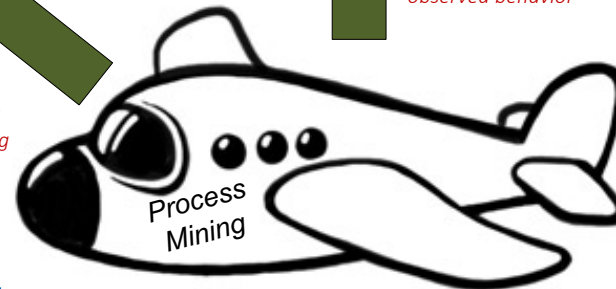
*avoiding
overfitting*

generalization

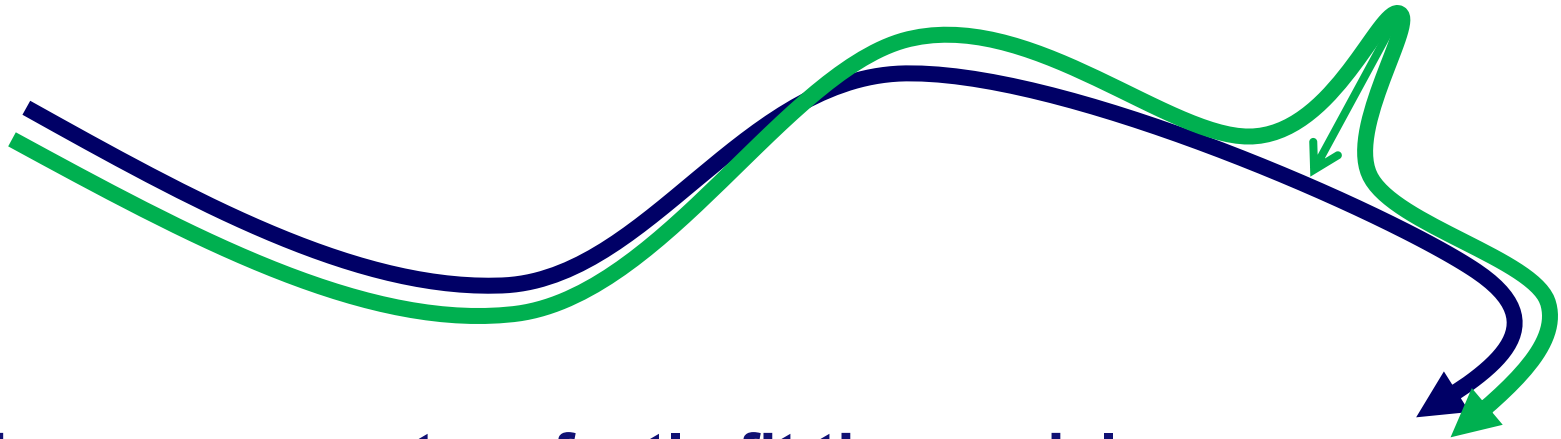
Occam's Razor

gravity

simplicity



When there are deviation we need to squeeze reality into the model



- Traces may not perfectly fit the model.
- Often we cannot throw non-fitting traces (loosing most of the data and/or introducing a hidden bias).
- Conformance checking techniques help us to map traces onto the "nearest" path in the model.

a	c	$\gg$	d	$\gg$	f	$\gg$
a	c	b	d	τ	$\gg$	h
$t1$	$t4$	$t3$	$t5$	$t7$		$t10$

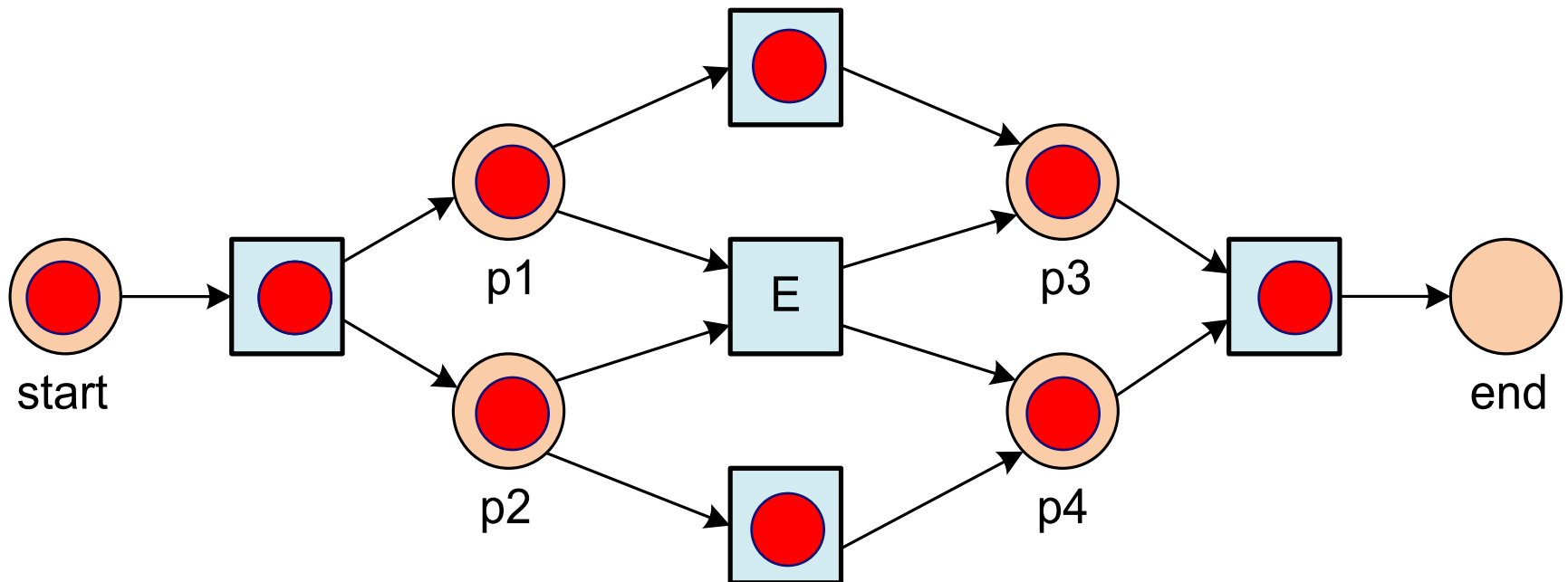
In the remainder we assume that through prepressing model and log are aligned



Assumption: every complete trace corresponds to a complete path through the model.

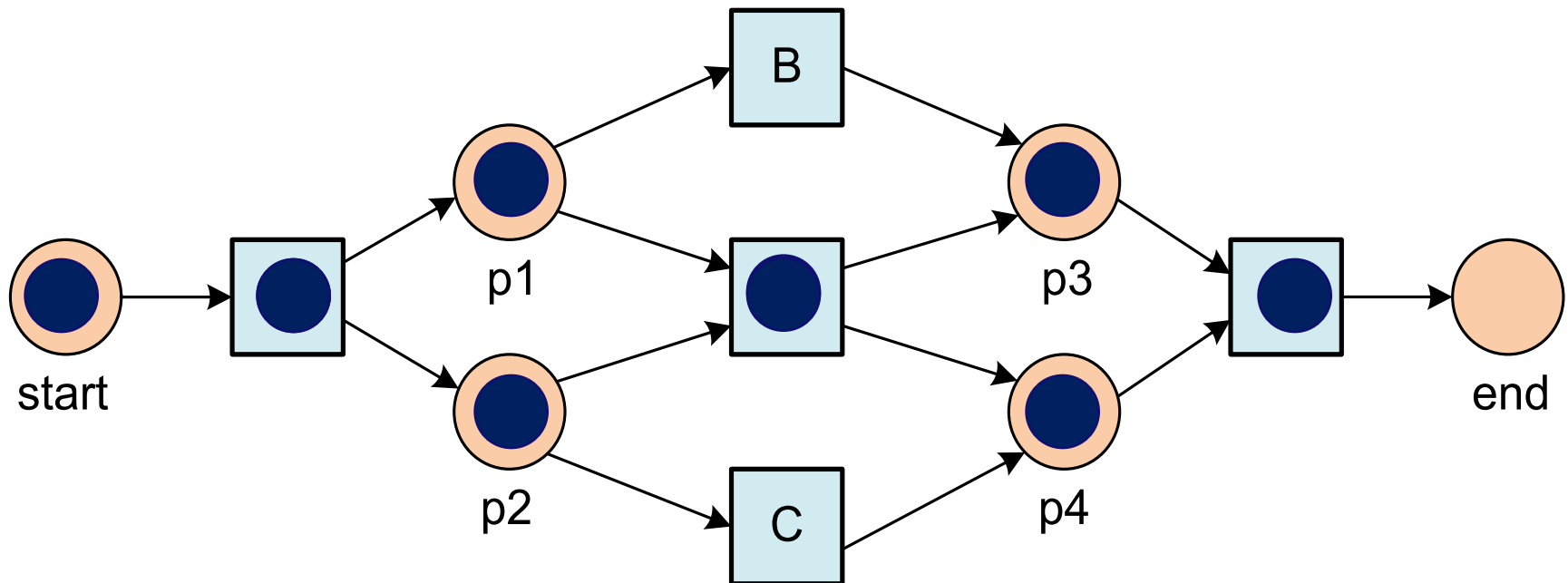
Decision mining: “Red” cases

A B C D



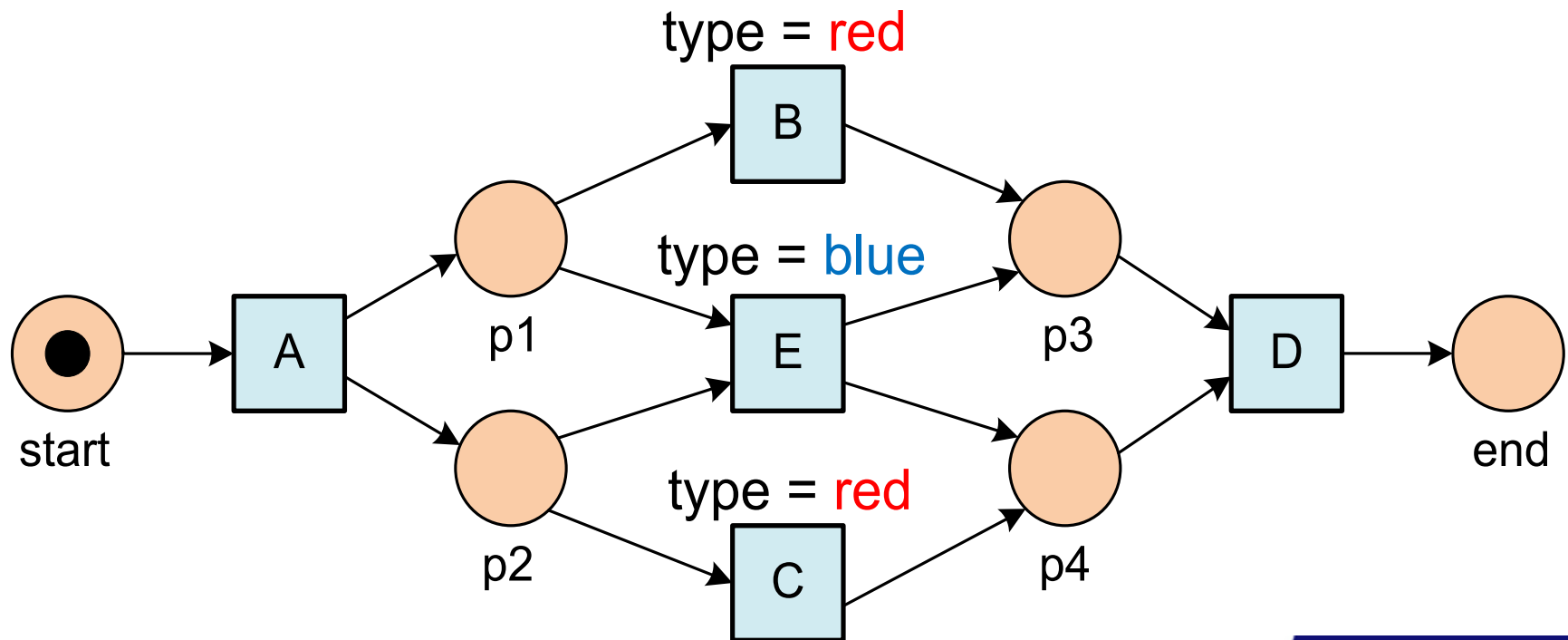
Decision mining: “Blue” cases

A E D

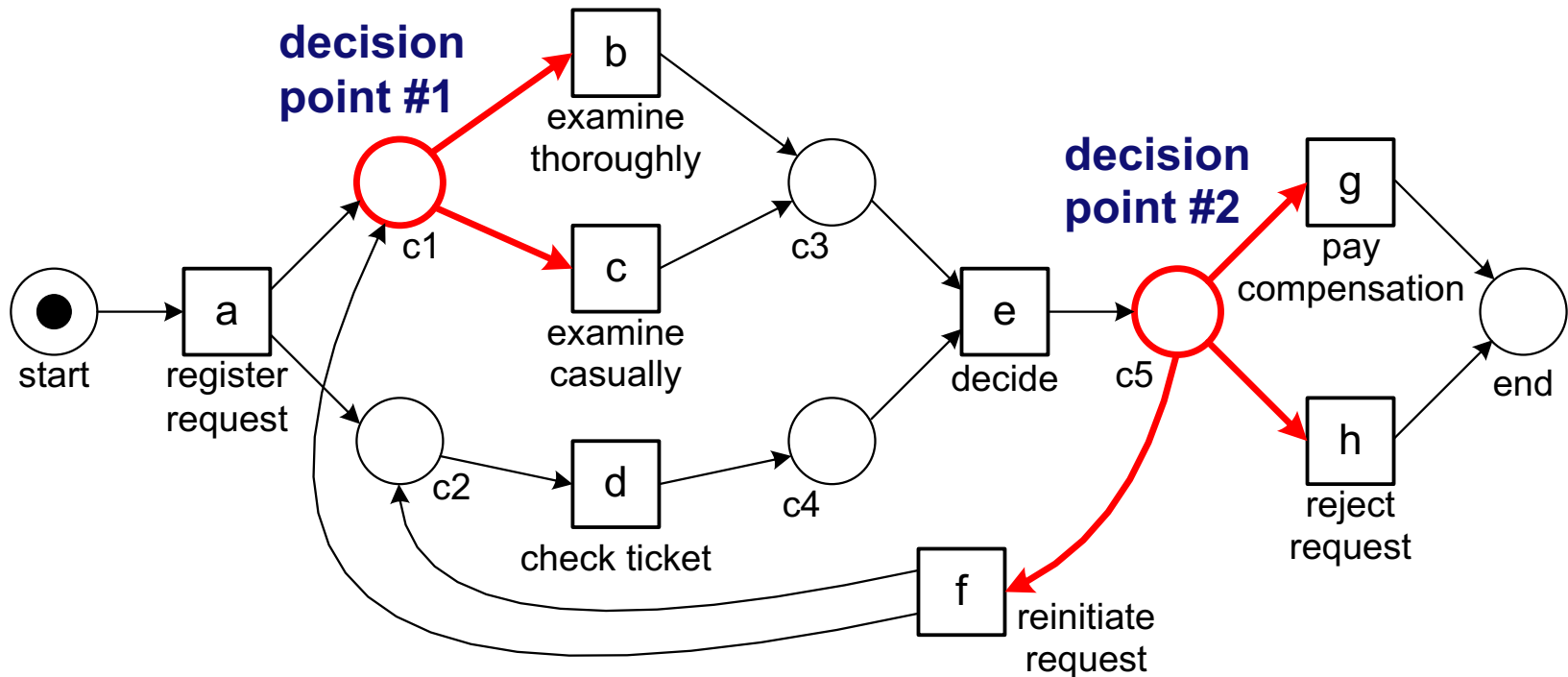


Guards ensure that the right path is taken

If red then B+C;
If blue then E;



Decision mining: places with multiple output arcs form decision points



In other notations often specific building blocks, e.g., XOR/OR gateways.

Remember:

classification using decision trees

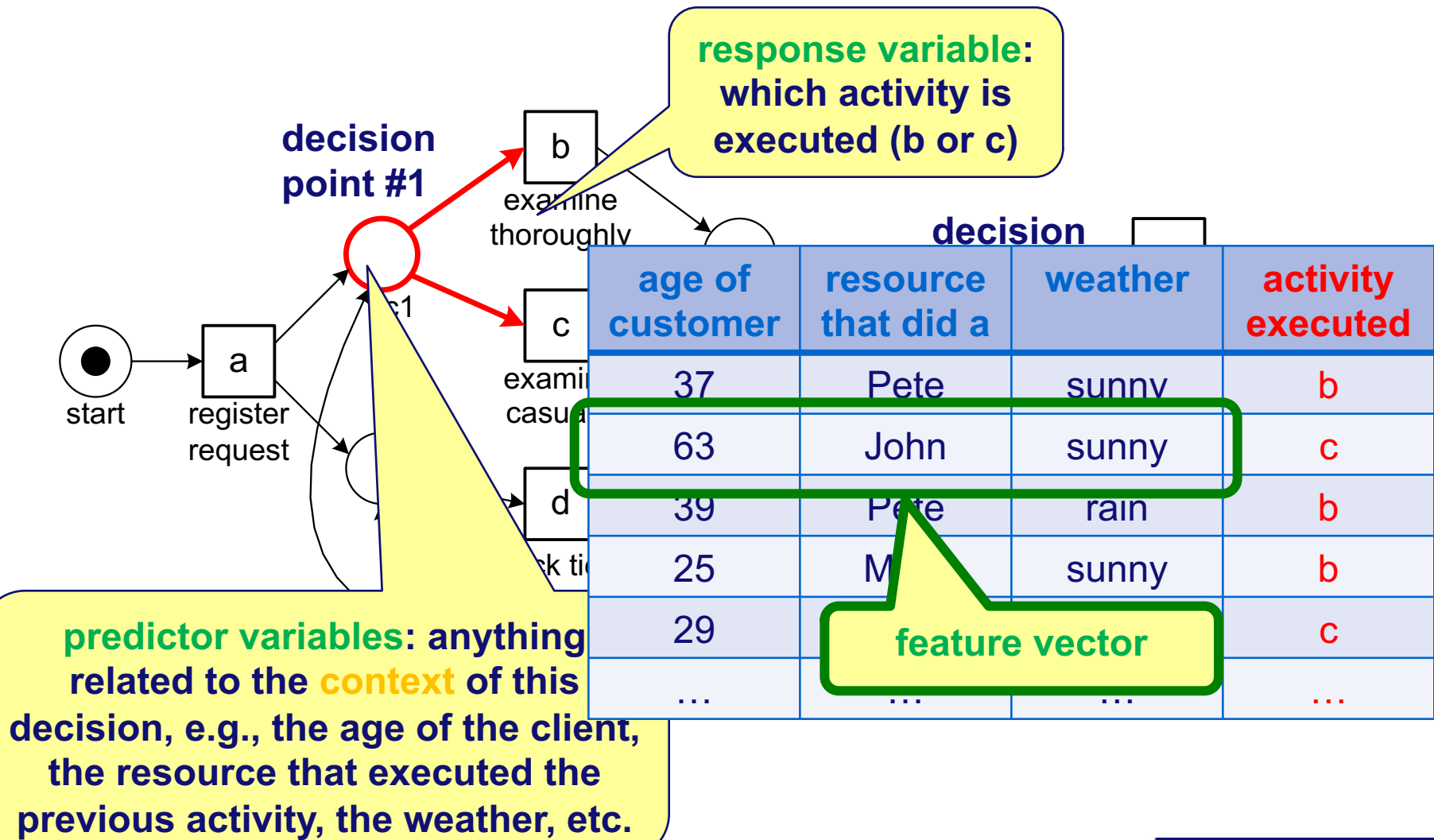
- Response variable (dependent variable): **claim** (yes/no).
- Predictor variables (independent variables): **gender, age, smoker, car brand.**



gender	age	smoker	car brand	claim
female	47	yes	Volvo	no
male	31	no	Alfa Romeo	yes
male	59	no	Alfa Romeo	yes
male	28	no	Fiat	no
male	44	no	BMW	no
female	27	no	Fiat	no
male	29	no	Subaru	no
male	44	yes	Subaru	yes
male	39	no	BMW	no
male	35	no	Subaru	yes

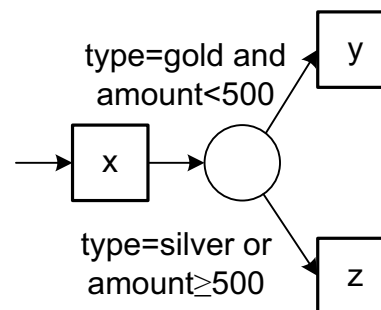
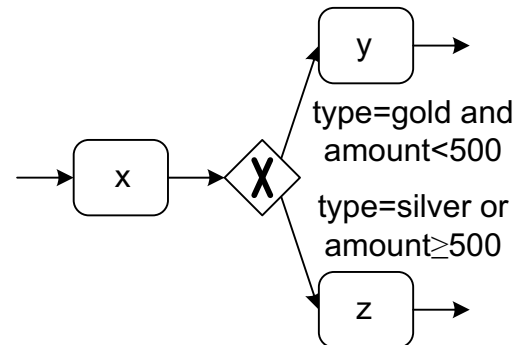
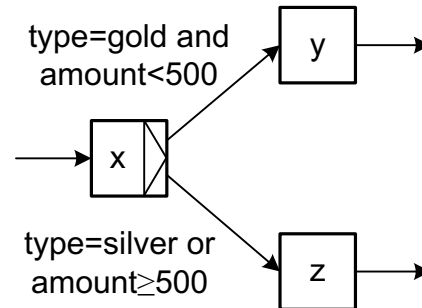
**Goal: explain
response variable
in terms of
relevant predictor
variables.**

Creating a classification problem



Example: XOR-split

type	region	amount	activity
gold	south	987.30	z
silver	north	178.70	z
gold	south	211.50	y
silver	west	587.70	z
silver	east	224.70	z
silver	south	278.50	z
gold	north	488.50	y
silver	west	443.20	z
silver	south	673.70	z
gold	west	413.50	y
silver	south	687.70	z
gold	south	987.30	z
silver	north	378.80	z
gold	south	314.50	y
silver	north	537.70	z
silver	west	158.70	z
gold	east	344.50	y
...	...	...	...

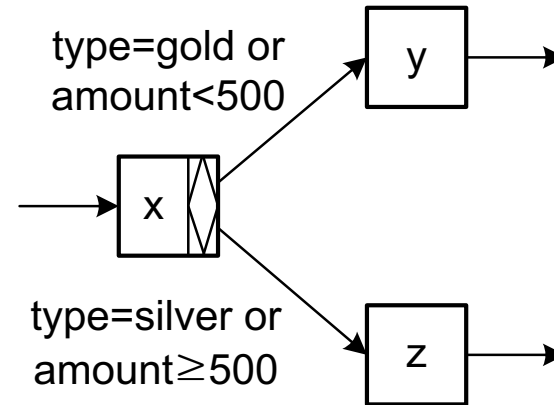


What are the **“features”** (predictor variables) influencing the decision?

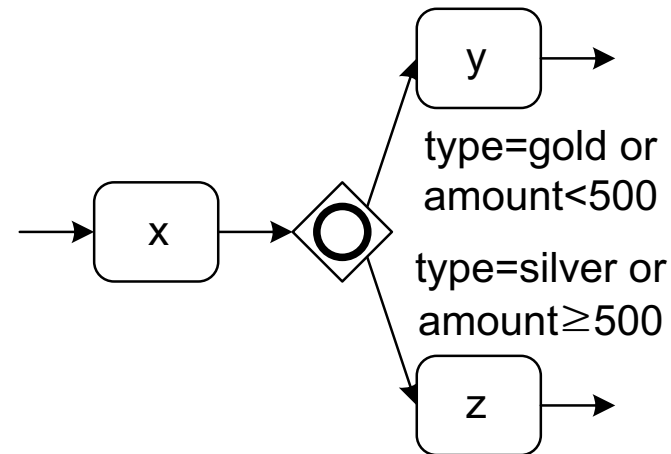
A **classification technique** like decision tree learning can be used to find such rules: **explain response variable** (dependent variable) **in terms of predictor variables** (independent variables).

Example: ~~X~~OR-split

type	region	amount	activity
gold	south	987.30	y and z
silver	north	178.70	y and z
gold	south	211.50	just y
silver	west	587.70	just z
silver	east	224.70	y and z
silver	south	278.50	y and z
gold	north	488.50	just y
silver	west	443.20	y and z
silver	south	673.70	just z
...	...	...	...

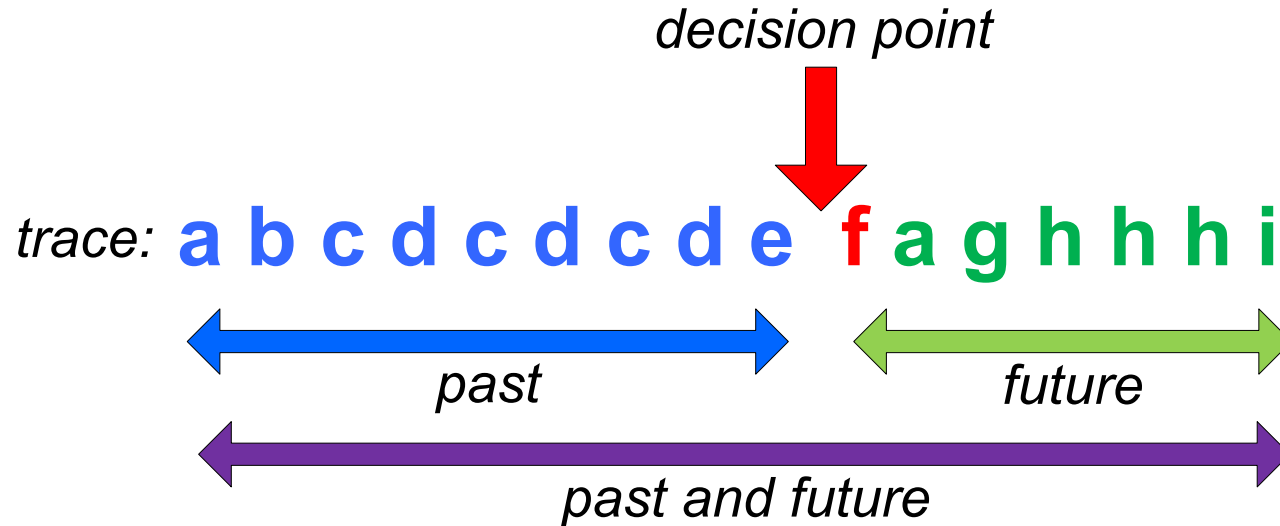


**Replay/
alignment will
tell what the
value of the
response
variable is.**



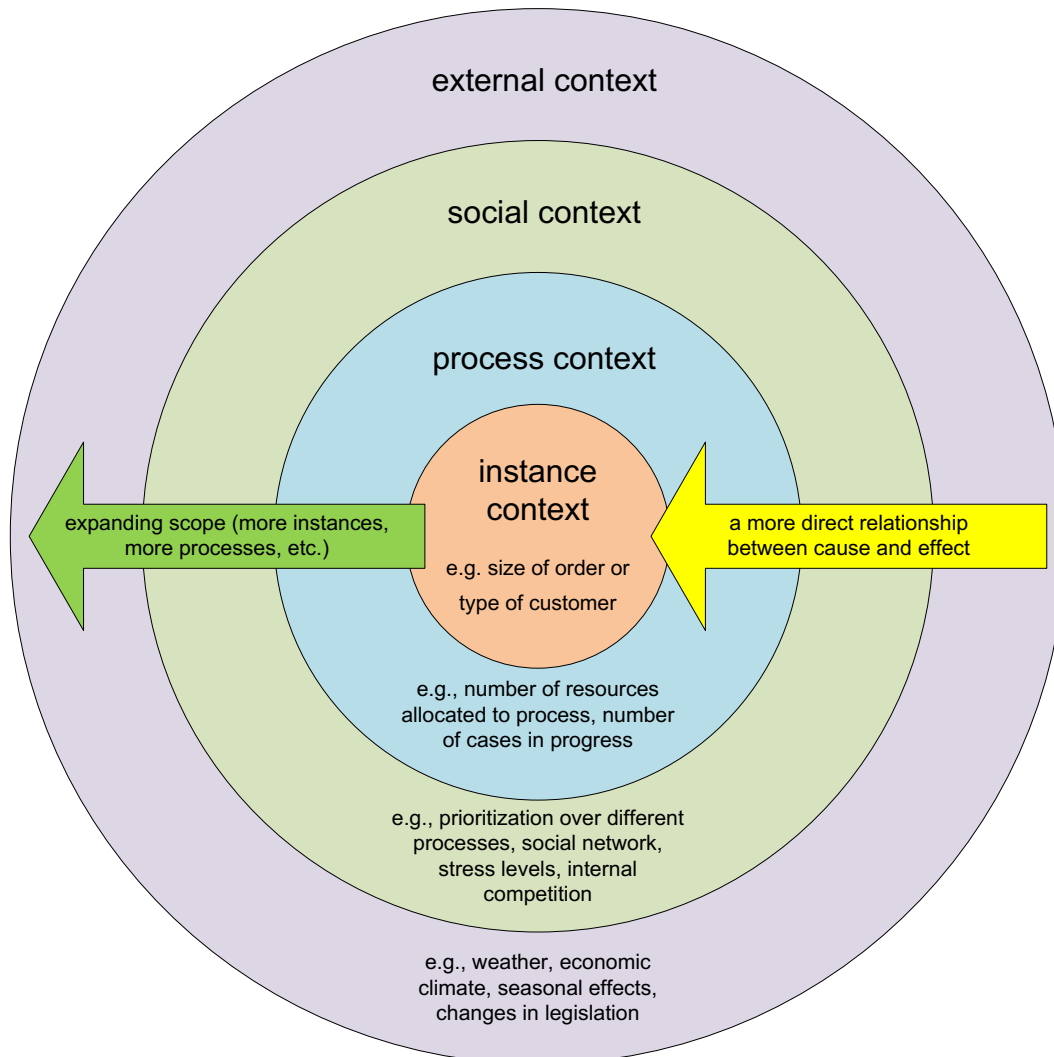
**Response variable (dependent variable) is now
an **activity set** rather than a single activity.**

Predictor variables based on process instance only



- Case variables
- Attributes (data, resource, etc.) of all past events
- Attributes (data, resource, etc.) of last event only
- Attributes (data, resource, etc.) of all events (including future).

Predictor variables based on context of the process instance

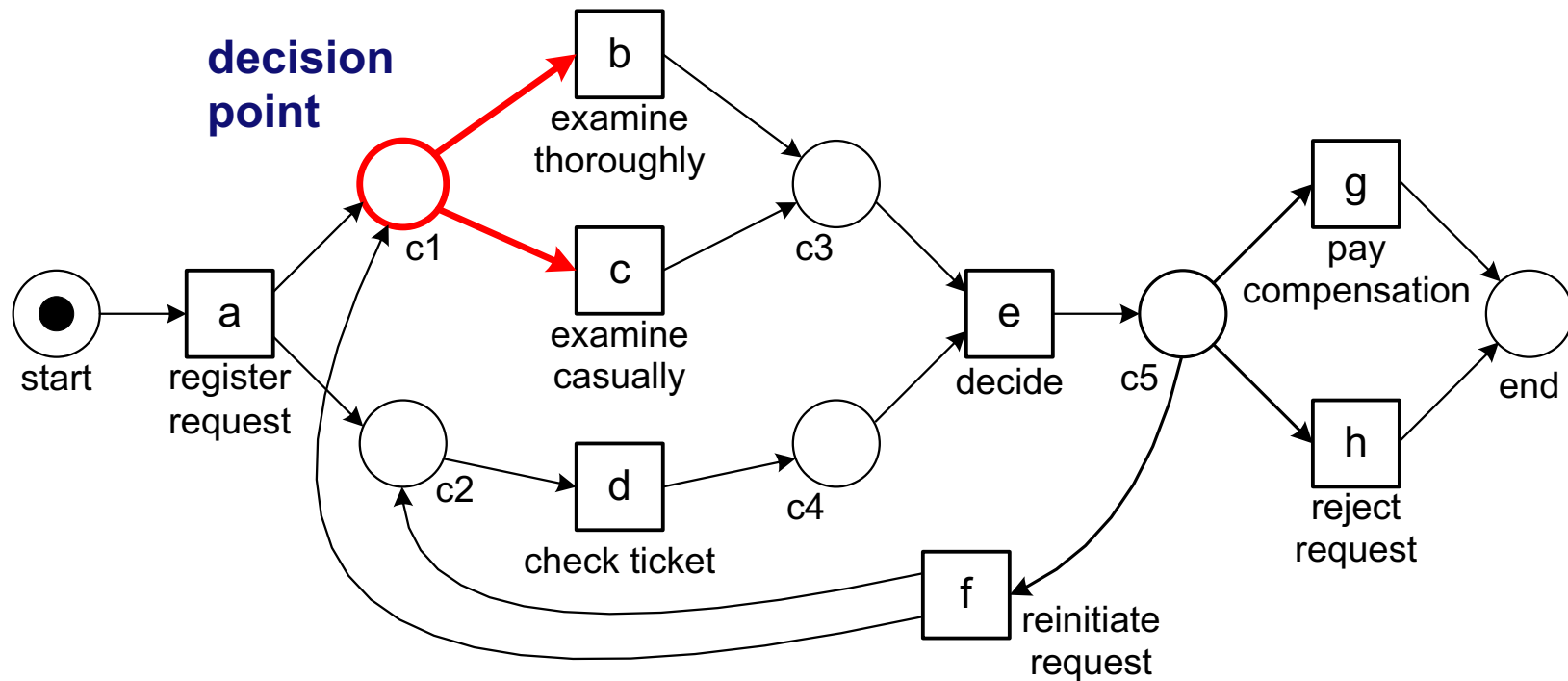


- Number of cases running (e.g. skip check if busy).
- Number of resources present.
- Day of the week.
- Weather.



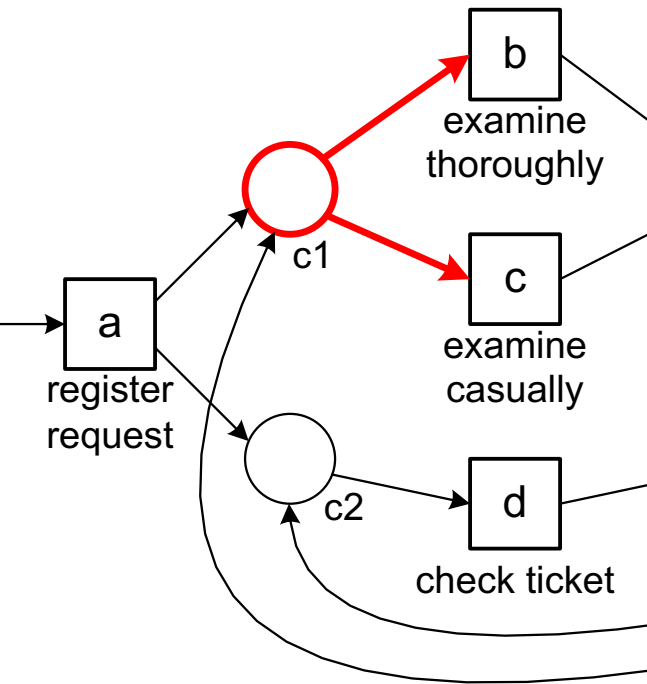
Problem: curse of dimensionality!

Question: Build classification problem to learn the decision point



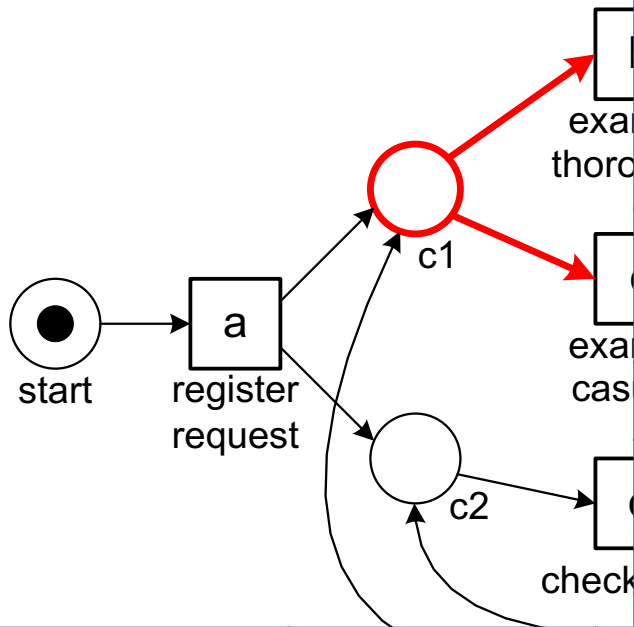
case	activity	resource	time	customer	amount
1	a	John	8.11	silver	500
2	a	Mary	8.12	gold	800
3	b	Sam	8.22	gold	800
4	c	Sam	8.22	gold	800
5	d	Sam	8.22	gold	800
6	e	Sam	8.22	gold	800
7	f	Sam	8.22	gold	800
8	g	Sam	8.22	gold	800
9	h	Sam	8.22	gold	800

Question: Build classification problem to learn the decision point



case	activity	resource	time	customer	amount
1	a	John	8.11	silver	500
2	a	Mary	8.12	gold	800
2	d	Sue	8.32	gold	800
1	b	John	9.12	silver	500
3	a	John	9.45	silver	300
3	c	Mary	9.56	silver	300
1	d	John	9.45	silver	500
2	c	Mary	9.56	gold	800
3	d	Mary	10.43	silver	300
4	a	John	11.34	gold	850
4	c	John	11.57	gold	850
...	...	...	...	...	...

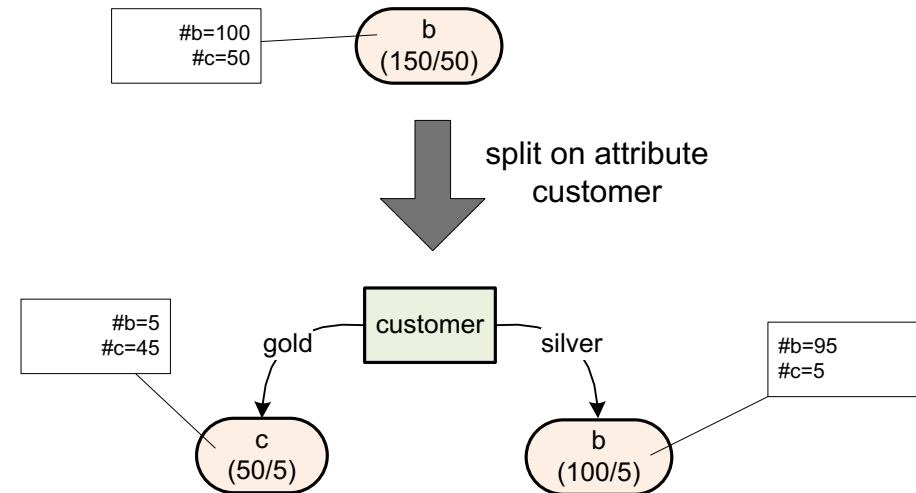
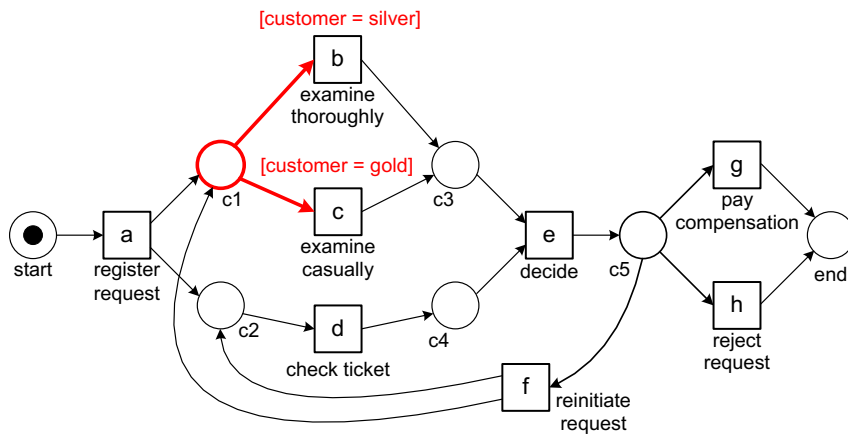
Classification problem (example, not all possible attributes included)



case	activity	resource	time	customer	amount
1	a	John	8.11	silver	500
2	a	Mary	8.12	gold	800
2	d	Sue	8.32	gold	800
1	b	John	9.12	silver	500
3	a	John	9.45	silver	300
3	c	Mary	9.56	silver	300
1	d	John	9.45	silver	500

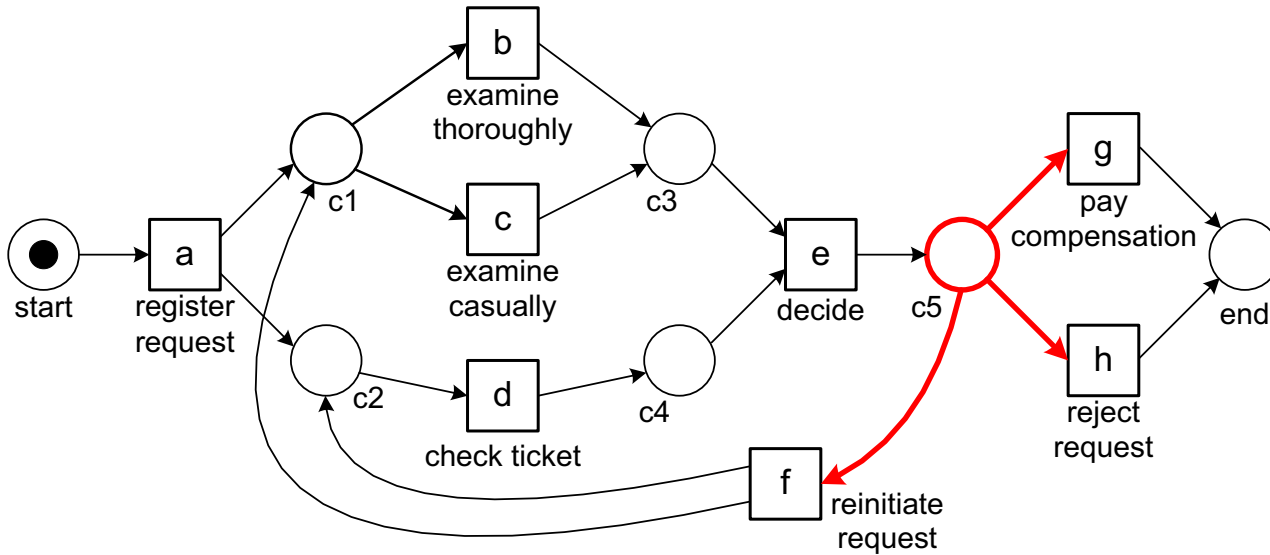
case	resource executing a	customer	amount	class		
1	John	silver	500	b	gold	850
2	Mary	gold	800	c	gold	850
3	John	silver	300	b	...	...
4	John	gold	850	c		

Decision tree and resulting guards

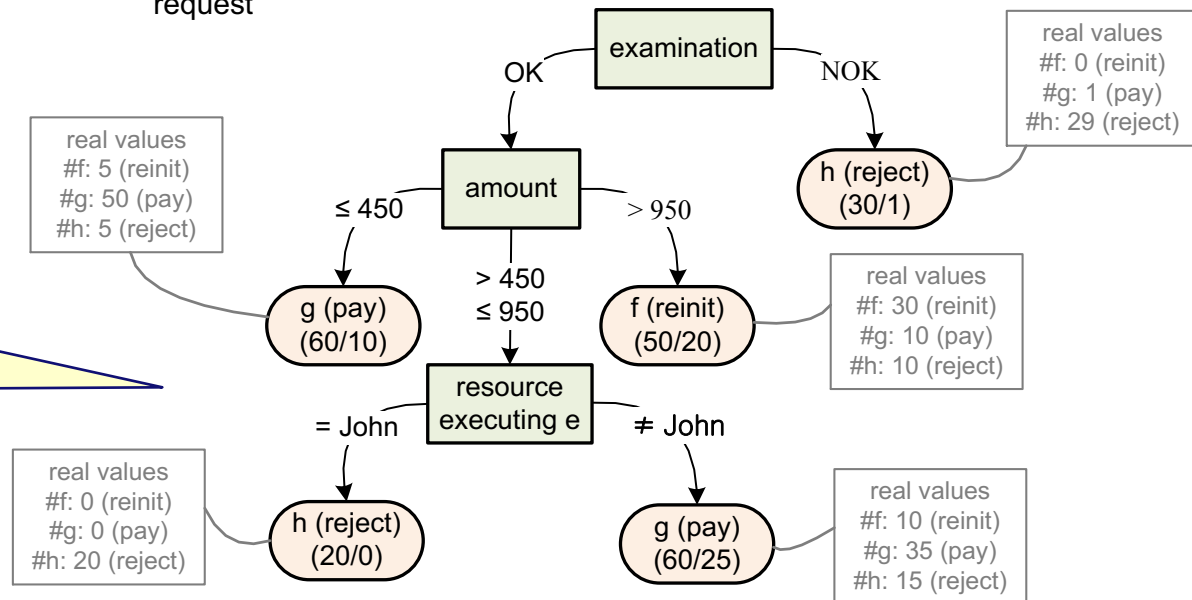


case	resource executing a	customer	amount	class
1	John	silver	500	b
2	Mary	gold	800	c
3	John	silver	300	b
4	John	gold	850	c

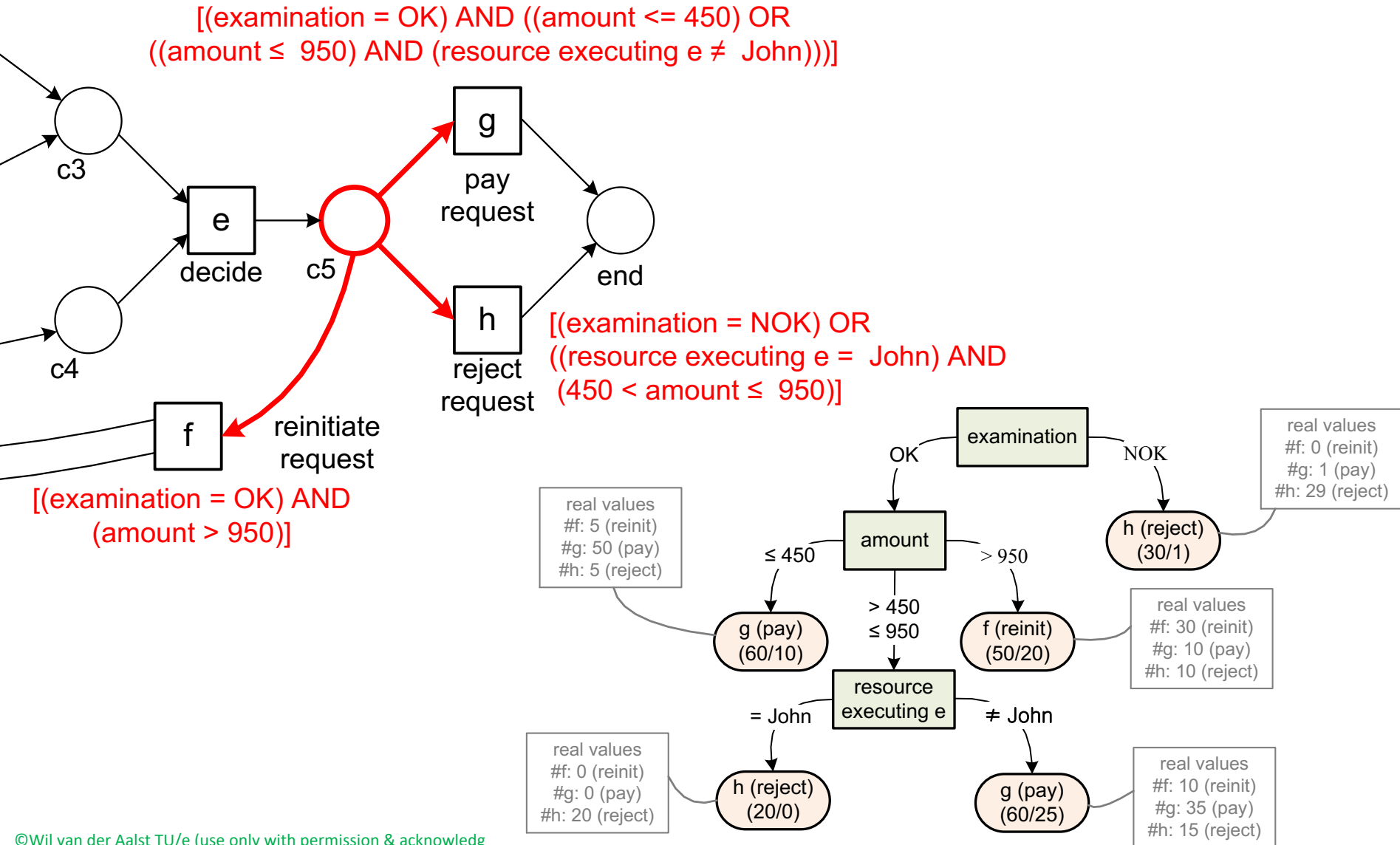
Question: create **guards** based on decision tree



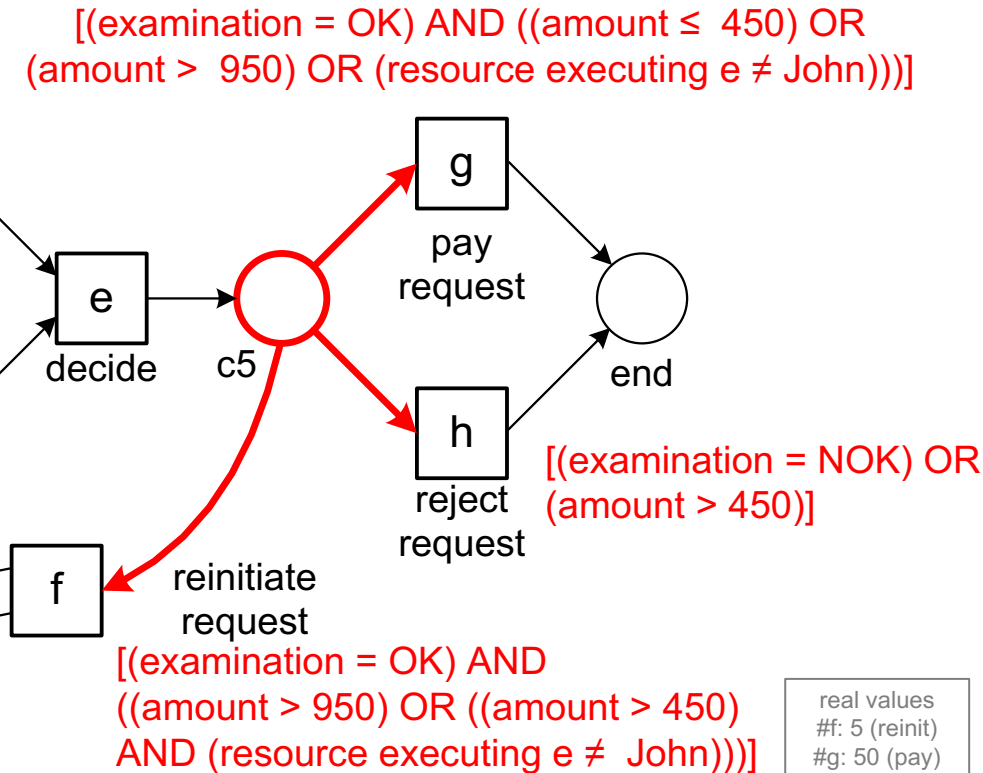
assume that this decision tree was learned for decision point



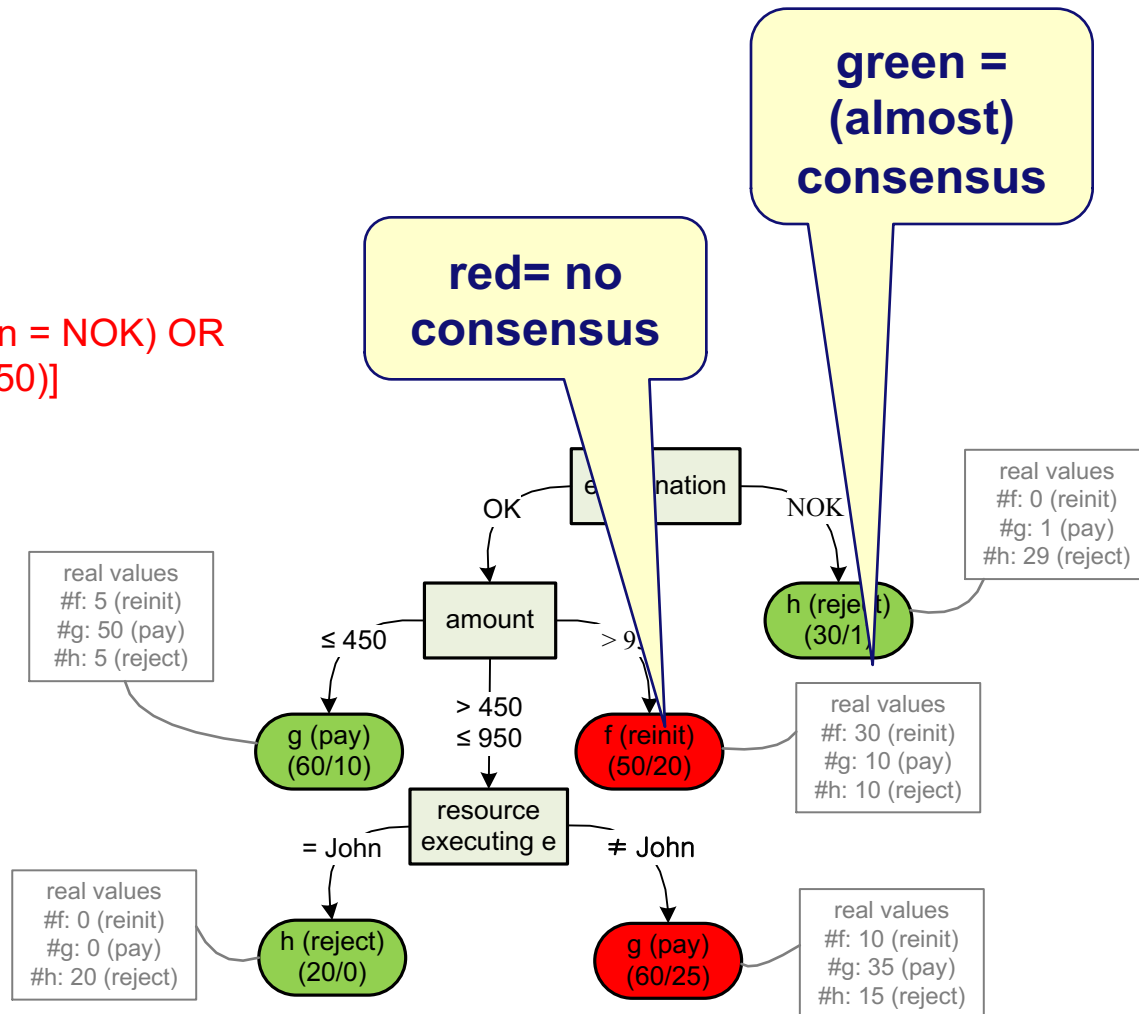
Deterministic guards



Non-deterministic guards



$[(\text{examination} = \text{OK}) \text{ AND } ((\text{amount} > 950) \text{ OR } ((\text{amount} > 450) \text{ AND } (\text{resource executing } e \neq \text{John})))]$



Data-dependent probabilities rather than guards

p_1 = if examination = NOK

then 1/30

else if amount $\leq$ 450

then 50/60

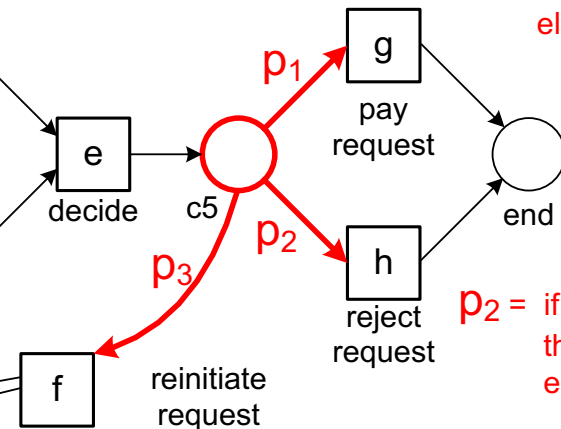
else if amount $>$ 950

then 10/50

else if resource executing a = John

then 0/20

else 35/60



p_2 = if examination = NOK

then 29/30

else if amount $\leq$ 450

then 5/60

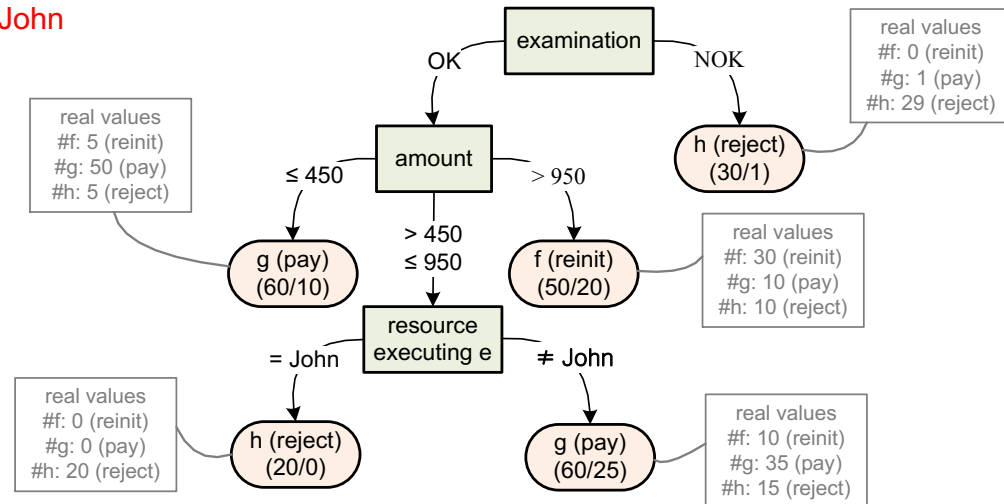
else if amount $>$ 950

then 10/50

else if resource executing a = John

then 20/20

else 15/60



p_3 = if examination = NOK

then 0/30

else if amount $\leq$ 450

then 5/60

else if amount $>$ 950

then 30/50

else if resource executing a = John

then 0/20

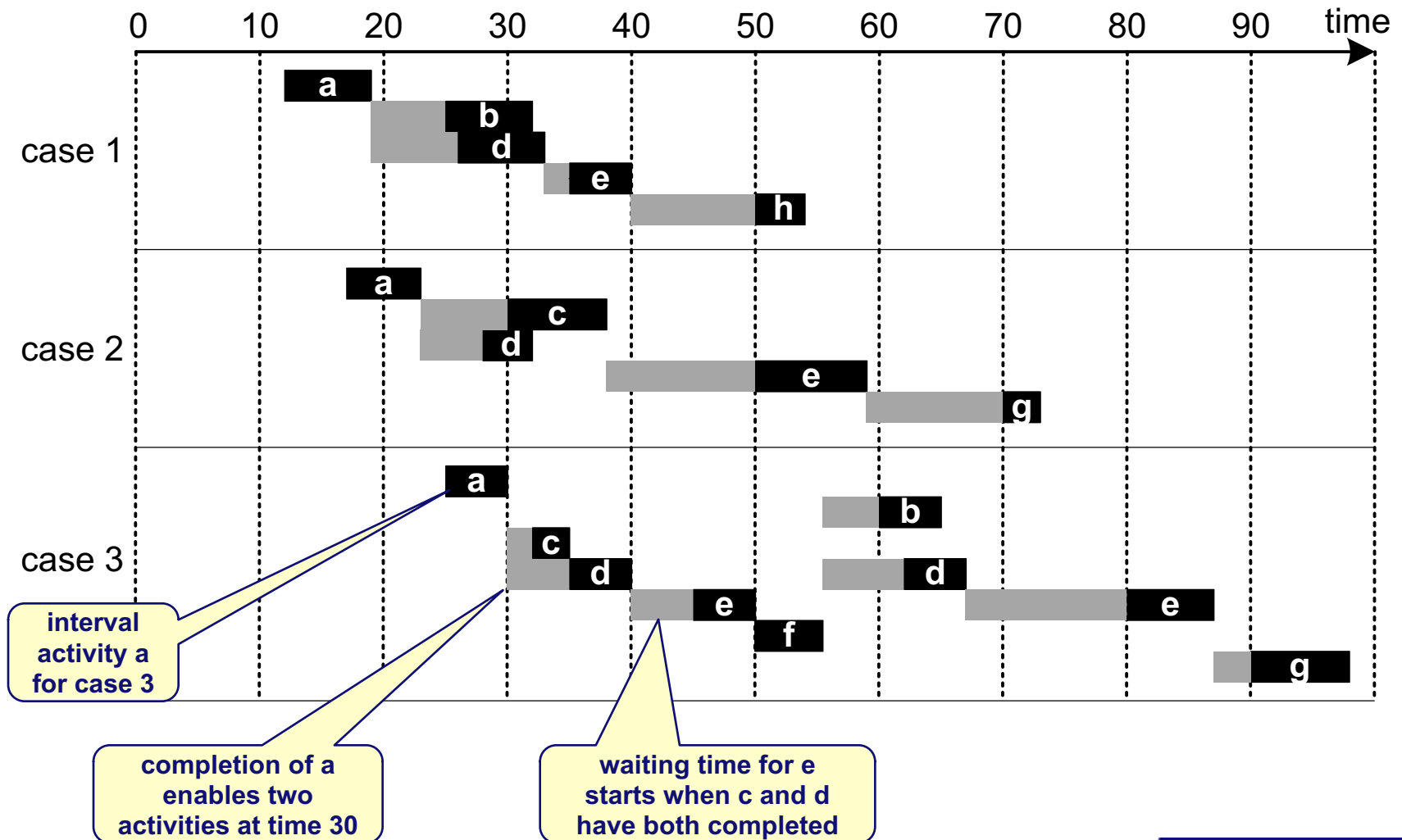
else 10/60

Learning time and probabilities

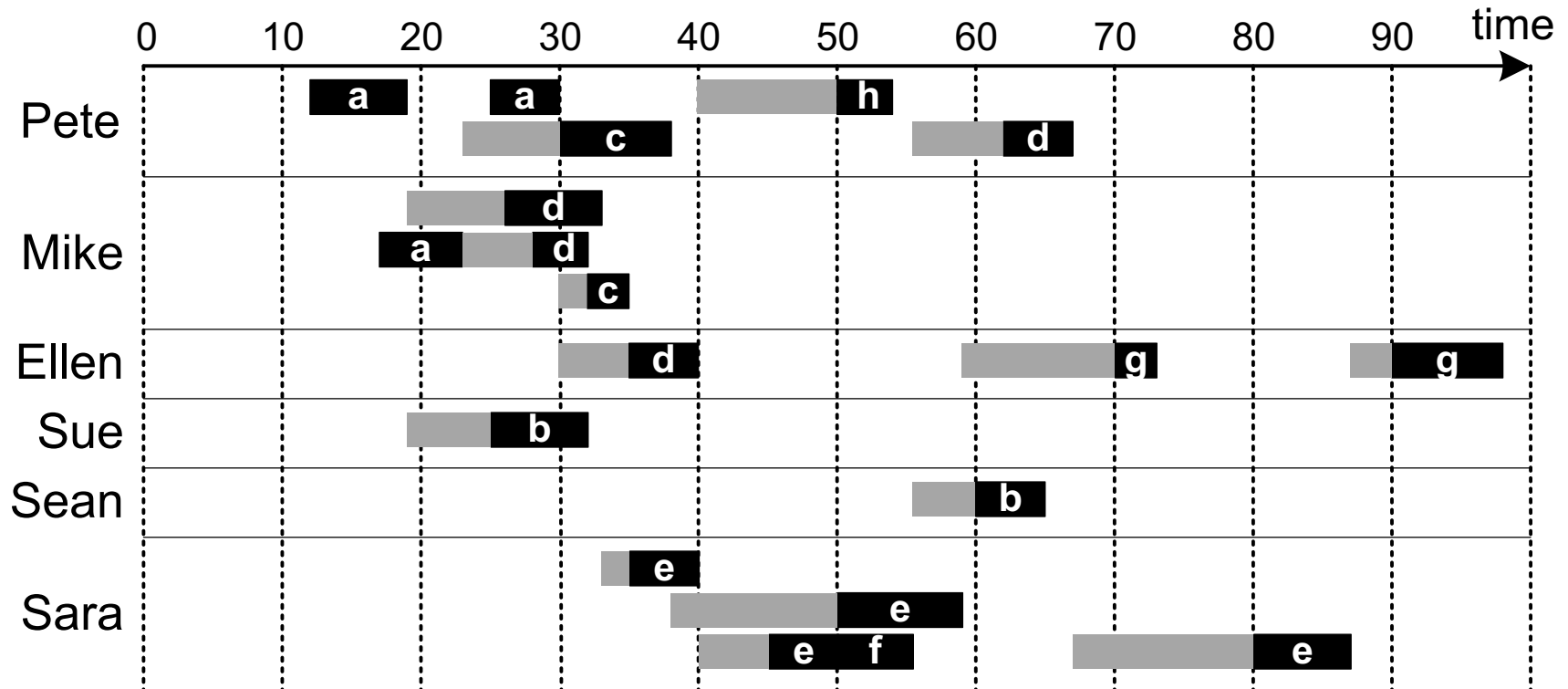
case id	trace
1	$\langle a_{start}^{12}, a_{complete}^{19}, b_{start}^{25}, d_{start}^{26}, b_{complete}^{32}, d_{complete}^{33}, e_{start}^{35}, e_{complete}^{40}, h_{start}^{50}, h_{complete}^{54} \rangle$
2	$\langle a_{start}^{17}, a_{complete}^{23}, d_{start}^{28}, c_{start}^{30}, d_{complete}^{32}, c_{complete}^{38}, e_{start}^{50}, e_{complete}^{59}, g_{start}^{70}, g_{complete}^{73} \rangle$
3	$\langle a_{start}^{25}, a_{complete}^{30}, c_{start}^{32}, c_{complete}^{35}, d_{start}^{35}, d_{complete}^{40}, e_{start}^{45}, e_{complete}^{50}, f_{start}^{50}, f_{complete}^{55}, b_{start}^{60}, d_{start}^{62}, b_{complete}^{65}, d_{complete}^{67}, e_{start}^{80}, e_{complete}^{87}, g_{start}^{90}, g_{complete}^{98} \rangle$
...	...

- Replay, as before, but now considering **timestamps**.
- Let us replay the first three cases in the event log:
 - **case 1** starts at time **12** and ends at time **54**,
 - **case 2** starts at time **17** and ends at time **73**,
 - **case 3** starts at time **25** and ends at time **98**.

Another view on the timed replay of the first three cases

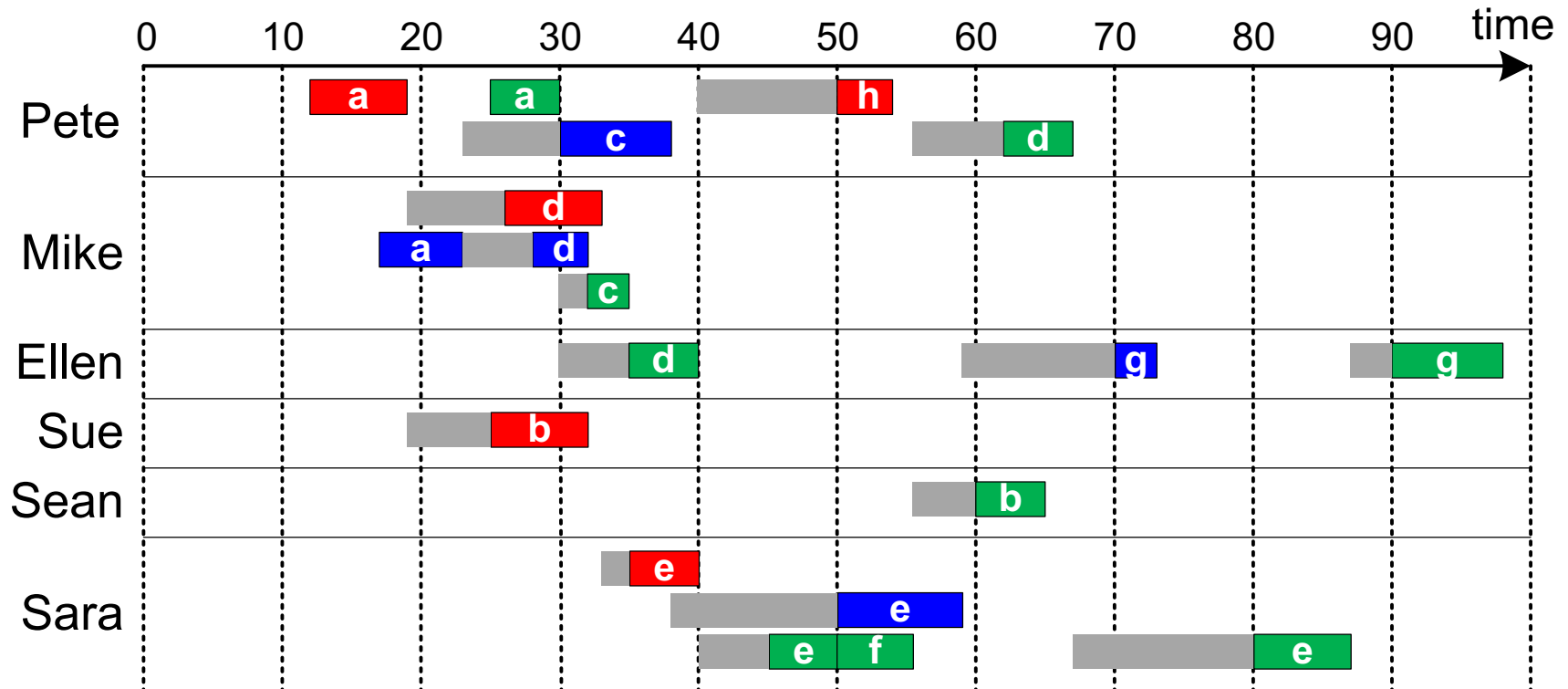


Timed replay projected onto resources



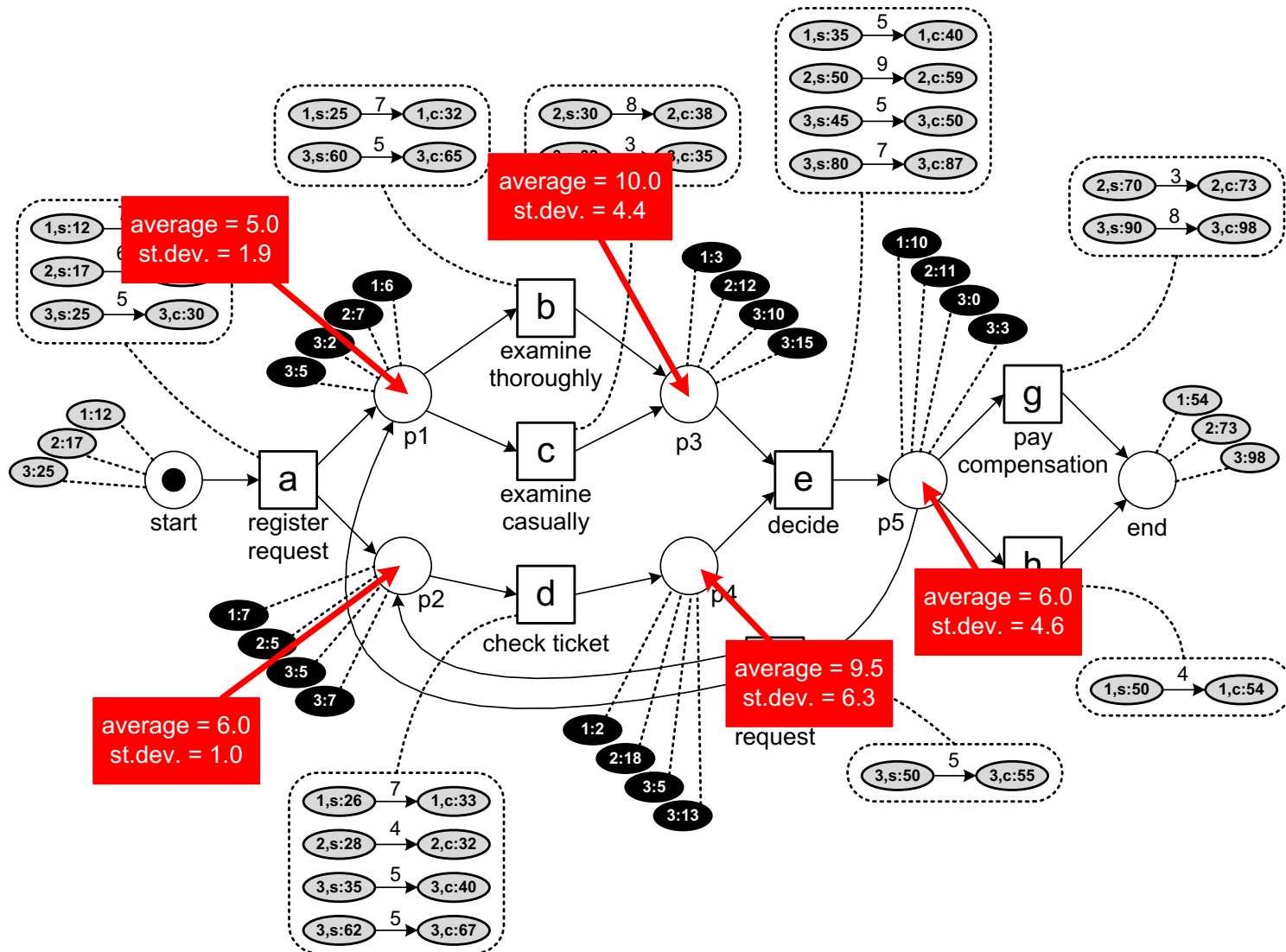
Timed replay projected onto resources

(activities colored by case)

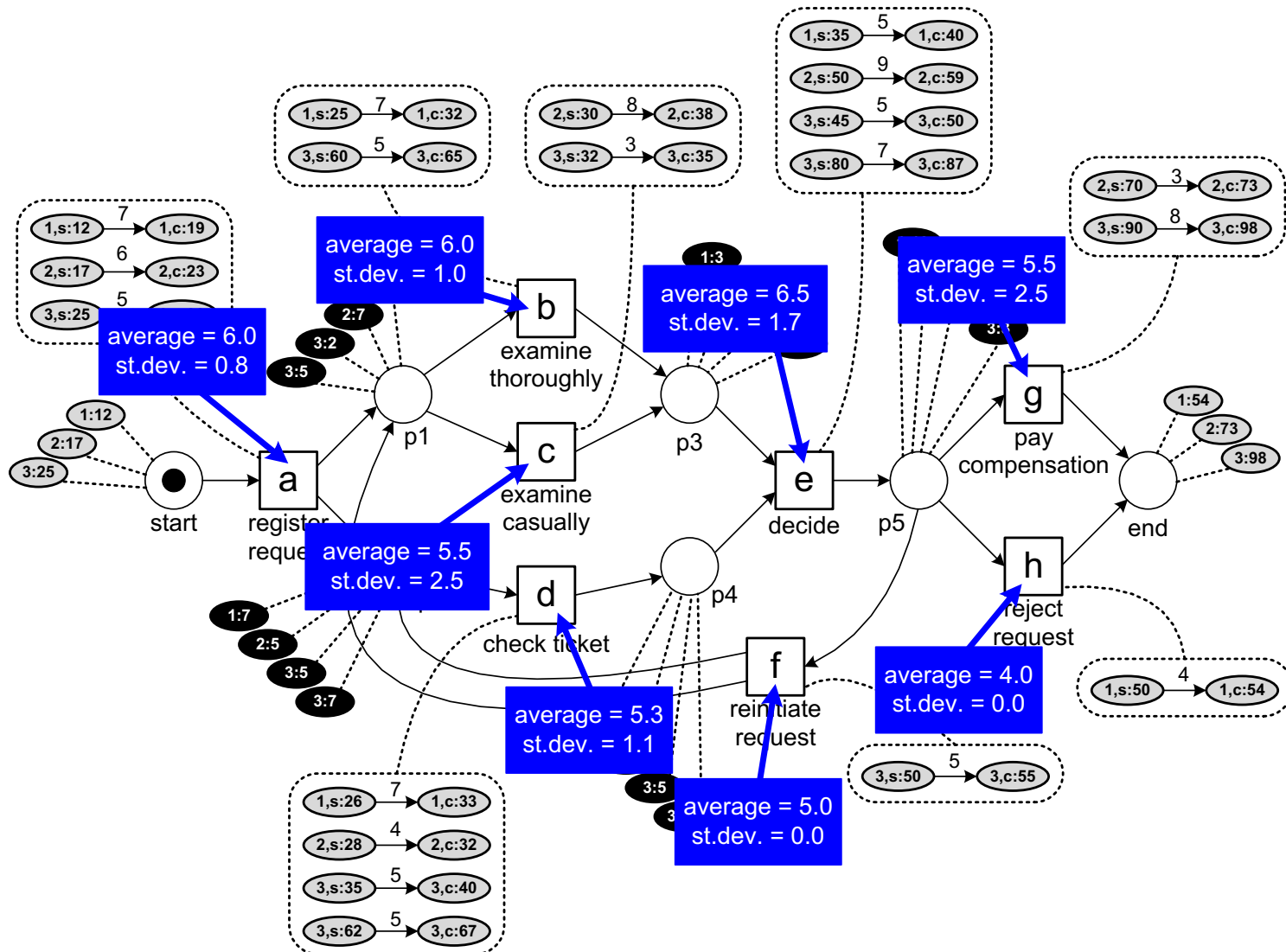


case id	trace
1	$\langle a_{start}^{12}, a_{complete}^{19}, b_{start}^{25}, d_{start}^{26}, b_{complete}^{32}, d_{complete}^{33}, e_{start}^{35}, e_{complete}^{40}, h_{start}^{50}, h_{complete}^{54} \rangle$
2	$\langle a_{start}^{17}, a_{complete}^{23}, d_{start}^{28}, c_{start}^{30}, d_{complete}^{32}, c_{complete}^{38}, e_{start}^{50}, e_{complete}^{59}, g_{start}^{70}, g_{complete}^{73} \rangle$
3	$\langle a_{start}^{25}, a_{complete}^{30}, c_{start}^{32}, c_{complete}^{35}, d_{start}^{40}, d_{complete}^{45}, e_{start}^{50}, e_{complete}^{55}, f_{start}^{50}, f_{complete}^{55}, b_{start}^{60}, b_{complete}^{62}, d_{start}^{65}, d_{complete}^{67}, e_{start}^{80}, e_{complete}^{87}, g_{start}^{90}, g_{complete}^{98} \rangle$
...	...

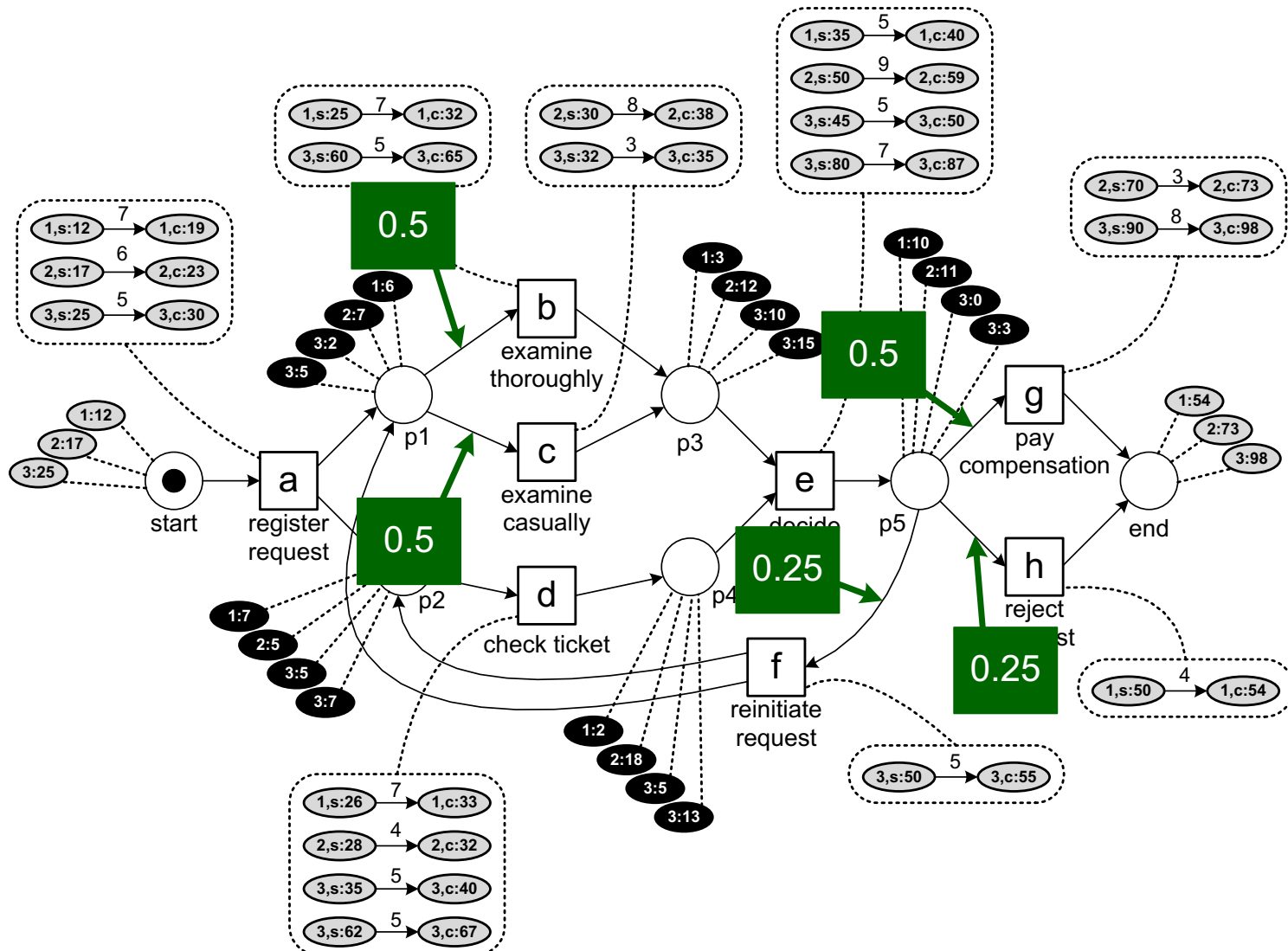
Waiting times



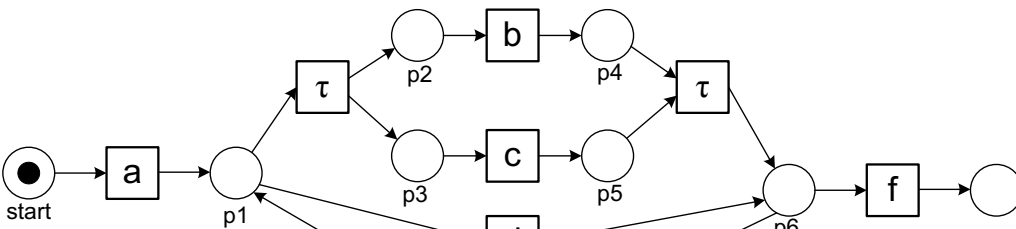
Service times



Routing probabilities



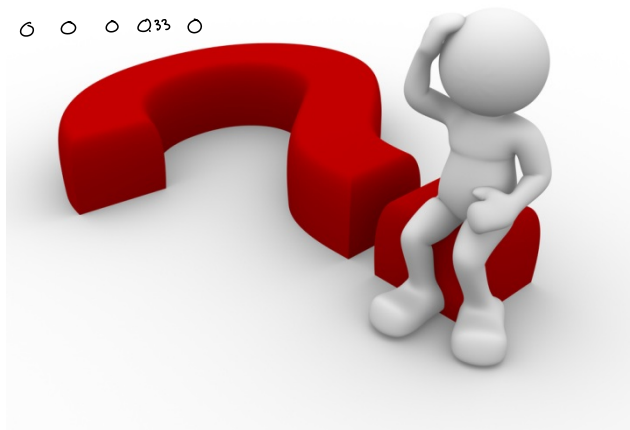
Estimate service times, waiting times, and routing probabilities



The values are given by:
 • Is "Pete" doing the activity "a"?
 • If no → 0, if yes → $\frac{\text{number of traces with "a" inside done by him}}{\text{number of traces in total}}$

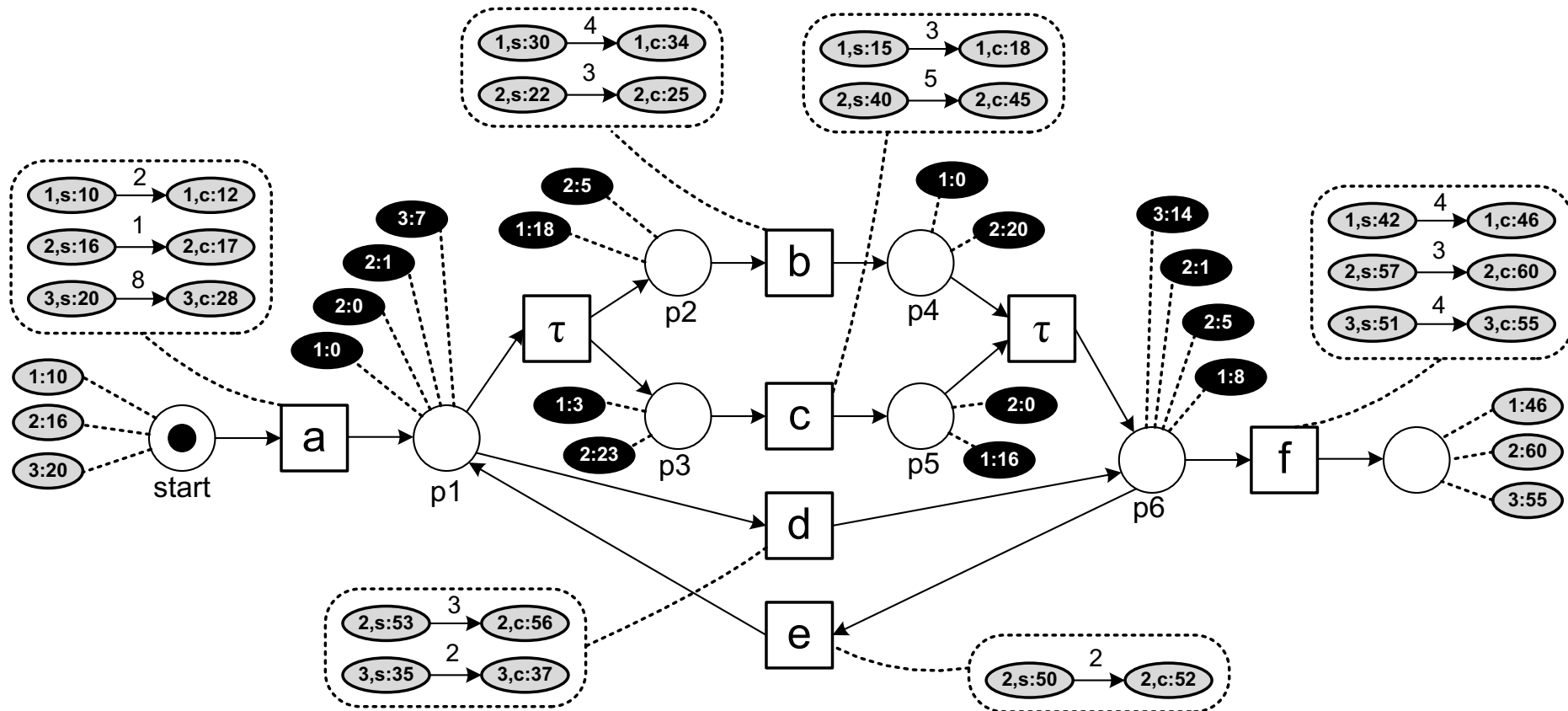
	a	b	c	d	e	f
Pete	1	0	0	0	0	0
Sue	0	0	0.67	0	0	0
Mary	0	0.67	0	0.67	0	0
Carol	0	0	0	0	0	1
Kirsten	0	0	0	0	0.33	0

id	trace
1	a, c, b, f
2	a, b, c, e, d, f
3	a, d, f



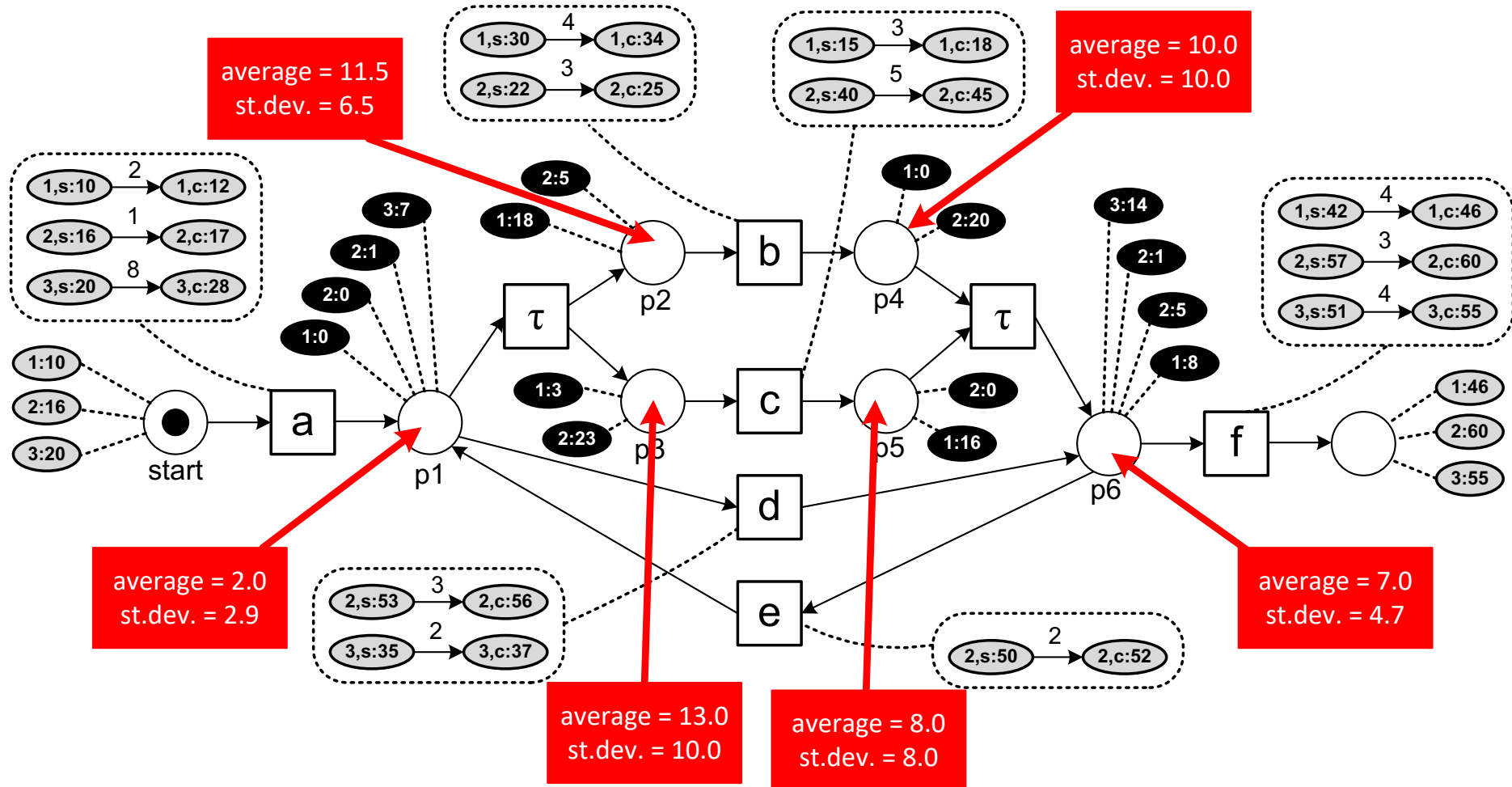
case id	activity	type	time	resource
1	a	start	10	Pete
1	a	complete	12	Pete
1	c	start	15	Sue
2	a	start	16	Pete
2	a	complete	17	Pete
1	c	complete	18	Sue
3	a	start	20	Pete
2	b	start	22	Mary
2	b	complete	25	Mary
3	a	complete	28	Pete
1	b	start	30	Mary
1	b	complete	34	Mary
3	d	start	35	Mary
3	d	complete	37	Mary
2	c	start	40	Sue
1	f	start	42	Carol
2	c	complete	45	Sue
1	f	complete	46	Carol
2	e	start	50	Kirsten
3	f	start	51	Carol
2	e	complete	52	Kirsten
2	d	start	53	Mary
3	f	complete	55	Carol
2	d	complete	56	Mary
2	f	start	57	Carol
2	f	complete	60	Carol

Times recorded during replay

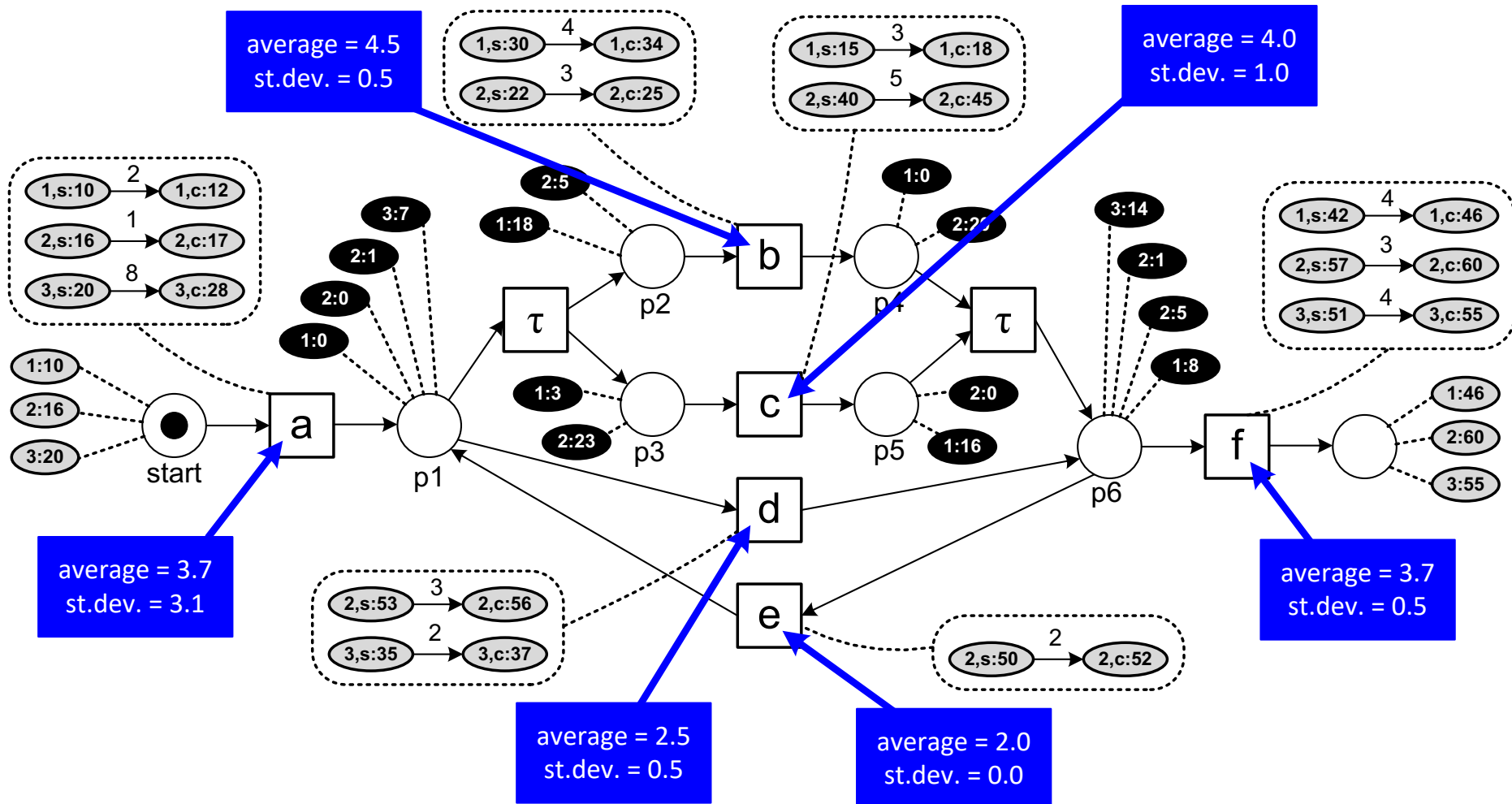


case id	activity	type	time	resource
1	a	start	10	Pete
1	a	complete	12	Pete
1	c	start	15	Sue
2	a	start	16	Pete
2	a	complete	17	Pete

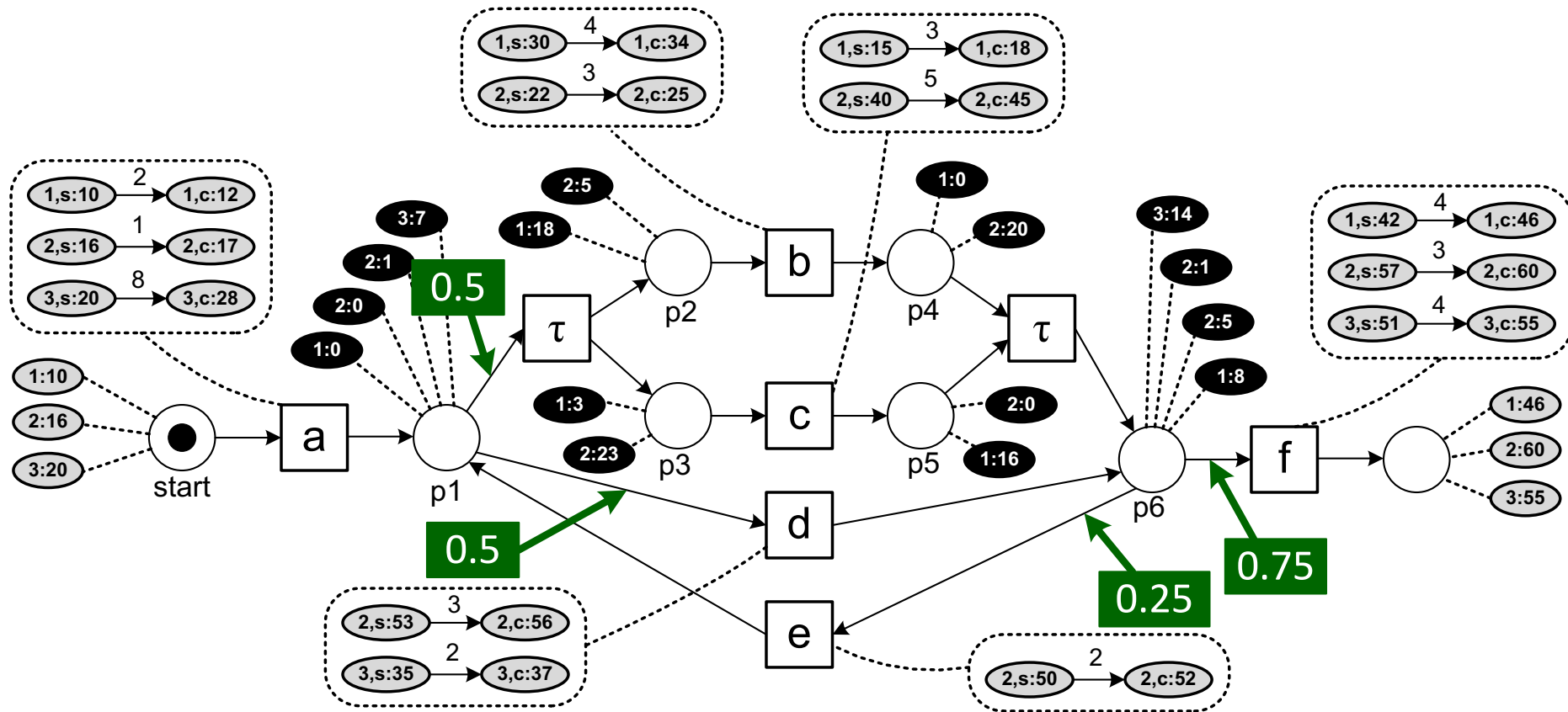
Waiting times



Service times



Routing probabilities



Organizational mining

case id trace

- 1 $\langle a^{Pete}, b^{Sue}, d^{Mike}, e^{Sara}, h^{Pete} \rangle$
 - 2 $\langle a^{Mike}, d^{Mike}, c^{Pete}, e^{Sara}, g^{Ellen} \rangle$
 - 3 $\langle a^{Pete}, c^{Mike}, d^{Ellen}, e^{Sara}, f^{Sara}, b^{Sean}, d^{Pete}, e^{Sara}, g^{Ellen} \rangle$
 - 4 $\langle a^{Pete}, d^{Mike}, b^{Sean}, e^{Sara}, h^{Ellen} \rangle$
 - 5 $\langle a^{Ellen}, c^{Mike}, d^{Pete}, e^{Sara}, f^{Sara}, d^{Ellen}, c^{Mike}, e^{Sara}, f^{Sara}, b^{Sue}, d^{Pete}, e^{Sara}, h^{Mike} \rangle$
 - 6 $\langle a^{Mike}, c^{Ellen}, d^{Mike}, e^{Sara}, g^{Mike} \rangle$
 -
-

(a = register request, b = examine thoroughly, c = examine casually, d = check ticket, e = decide, f = reinitiate request, g = pay compensation, and h = reject request)

Resource-activity matrix

mean number of times a resource performs an activity per case

	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>	<i>h</i>
Pete	0.3	0	0.345	0.69	0	0	0.135	0.165
Mike	0.5	0	0.575	1.15	0	0	0.225	0.275
Ellen	0.2	0	0.23	0.46	0	0	0.09	0.11
Sue	0	0.46	0	0	0	0	0	0
Sean	0	0.69	0	0	0	0	0	0
Sara	0	0	0	0	2.3	1.3	0	0

Activity a is executed exactly once for each case (hence the sum of the first column is 1). Pete, Mike, and Ellen are the only ones executing this activity. In 30% of the cases, a is executed by Pete, 50% is executed by Mike, and 20% is executed by Ellen. Activities e and f are always executed by Sara. Activity e is executed, on average, 2.3 times per case. Etc.

Create resource-activity matrix

Traces=

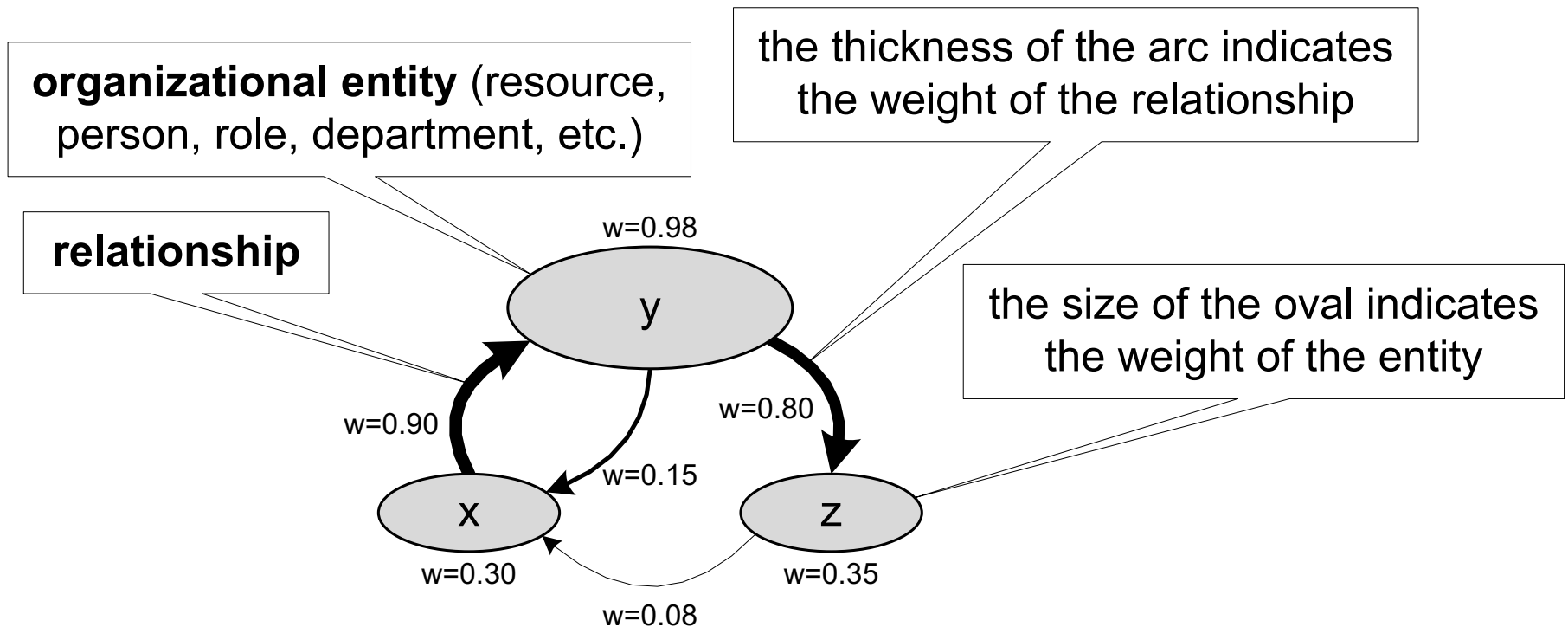
- 1: a c b f
- 2: a b c e d f
- 3: a d f

	a	b	c	d	e	f
Pete	3	0	0	0	0	0
Mary	0	2	0	2	0	0
Sue	0	0	3	0	0	0
Kirsten	0	0	0	0	1	0
Carol	0	0	0	0	0	3

	a	b	c	d	e	f
Pete	1.00	0.00	0.00	0.00	0.00	0.00
Mary	0.00	0.67	0.00	0.67	0.00	0.00
Sue	0.00	0.00	0.67	0.00	0.00	0.00
Kirsten	0.00	0.00	0.00	0.00	0.33	0.00
Carol	0.00	0.00	0.00	0.00	0.00	1.00

case id	activity	type	time	resource
1	a	start	10	Pete
1	a	complete	12	Pete
1	c	start	15	Sue
2	a	start	16	Pete
2	a	complete	17	Pete
1	c	complete	18	Sue
3	a	start	20	Pete
2	b	start	22	Mary
2	b	complete	25	Mary
3	a	complete	28	Pete
1	b	start	30	Mary
1	b	complete	34	Mary
3	d	start	35	Mary
3	d	complete	37	Mary
2	c	start	40	Sue
1	f	start	42	Carol
2	c	complete	45	Sue
1	f	complete	46	Carol
2	e	start	50	Kirsten
3	f	start	51	Carol
2	e	complete	52	Kirsten
2	d	start	53	Mary
3	f	complete	55	Carol
2	d	complete	56	Mary
2	f	start	57	Carol
2	f	complete	60	Carol

Social network



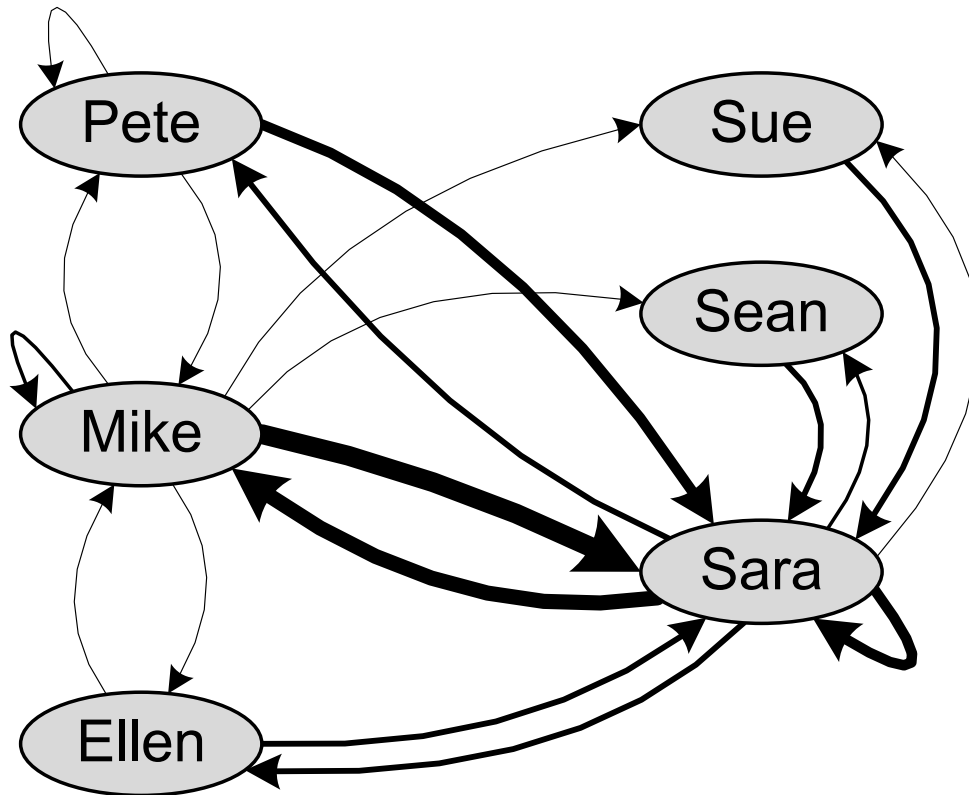
Handover of ~~work matrix~~

	Pete	Mike	Ellen	Sue	Sean	Sara
Pete	0.135	0.225	0.09	0.06	0.09	1.035
Mike	0.225	0.375	0.15	0.1	0.15	1.725
Ellen	0.09	0.15	0.06	0.04	0.06	0.69
Sue	0	0	0	0	0	0.46
Sean	0	0	0	0	0	0.69
Sara	0.885	1.475	0.59	0.26	0.39	1.3

Count the number of times work is handed over from one resource to another (on average per case).

The causal dependencies in the process model are used to count handovers in the event log.

Social network based on handover of work (threshold of 0.1)

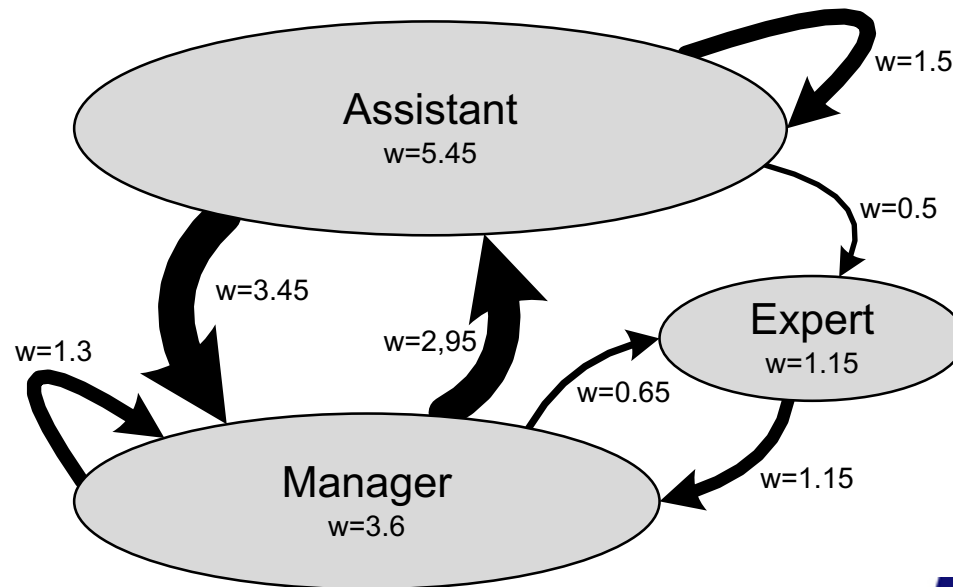


In this figure only the thickness of the arcs is based on frequencies.

	Pete	Mike	Ellen	Sue	Sean	Sara
Pete	0.135	0.225	0.09	0.06	0.09	1.035
Mike	0.225	0.375	0.15	0.1	0.15	1.725
Ellen	0.09	0.15	0.06	0.04	0.06	0.69
Sue	0	0	0	0	0	0.46
Sean	0	0	0	0	0	0.69
Sara	0.885	1.475	0.59	0.26	0.39	1.3

Handover of work at role level

	Assistant	Expert	Manager
Assistant	1.5	0.5	3.45
Expert	0	0	1.15
Manager	2.95	0.65	1.3



In this figure also the size of each node is based on frequencies.

Create handover of work matrix

case id	activity	type	time	resource
1	a	start	10	Pete
1	a	complete	12	Pete
1	c	start	15	Sue
2	a	start	16	Pete
2	a	complete	17	Pete
1	c	complete	18	Sue
3	a	start	20	Pete
2	b	start	22	Mary
2	b	complete	25	Mary
3	a	complete	28	Pete
1	b	start	30	Mary
1	b	complete	34	Mary
3	d	start	35	Mary
3	d	complete	37	Mary
2	c	start	40	Sue
1	f	start	42	Carol
2	c	complete	45	Sue
1	f	complete	46	Carol
2	e	start	50	Kirsten
3	f	start	51	Carol
2	e	complete	52	Kirsten
2	d	start	53	Mary
3	f	complete	55	Carol
2	d	complete	56	Mary
2	f	start	57	Carol
2	f	complete	60	Carol



- Count the number of times work is handed over from one resource to another (on average per case).
- Ignore process model, no need to consider causalities and concurrency.

Handover of work matrix

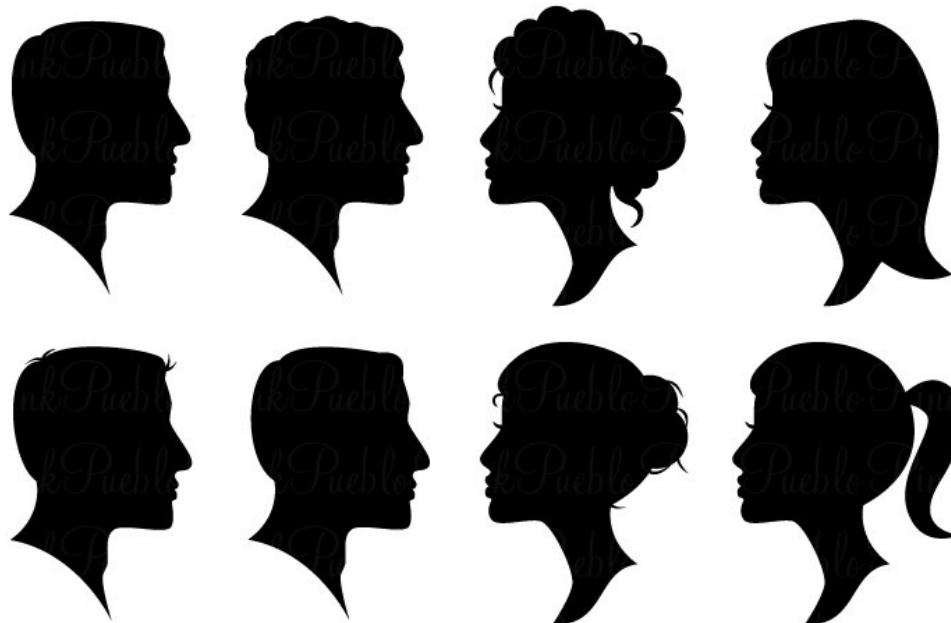
case id	activity	type	time	resource
1	a	start	10	Pete
1	a	complete	12	Pete
1	c	start	15	Sue
2	a	start	16	Pete
2	a	complete	17	Pete
1	c	complete	18	Sue
3	a	start	20	Pete
2	b	start	22	Mary
2	b	complete	25	Mary
3	a	complete	28	Pete
1	b	start	30	Mary
1	b	complete	34	Mary
3	d	start	35	Mary
3	d	complete	37	Mary
2	c	start	40	Sue
1	f	start	42	Carol
2	c	complete	45	Sue
1	f	complete	46	Carol
2	e	start	50	Kirsten
3	f	start	51	Carol
2	e	complete	52	Kirsten
2	d	start	53	Mary
3	f	complete	55	Carol
2	d	complete	56	Mary
2	f	start	57	Carol
2	f	complete	60	Carol

1. Pete, Sue, Mary, Carol
2. Pete, Mary, Sue, Kirsten, Mary, Carol
3. Pete, Mary, Carol

	Pete	Mary	Sue	Kirsten	Carol
Pete	0.00	0.66	0.33	0.00	0.00
Mary	0.00	0.00	0.33	0.00	1.00
Sue	0.00	0.33	0.00	0.33	0.00
Kirsten	0.00	0.33	0.00	0.00	0.00
Carol	0.00	0.00	0.00	0.00	0.00

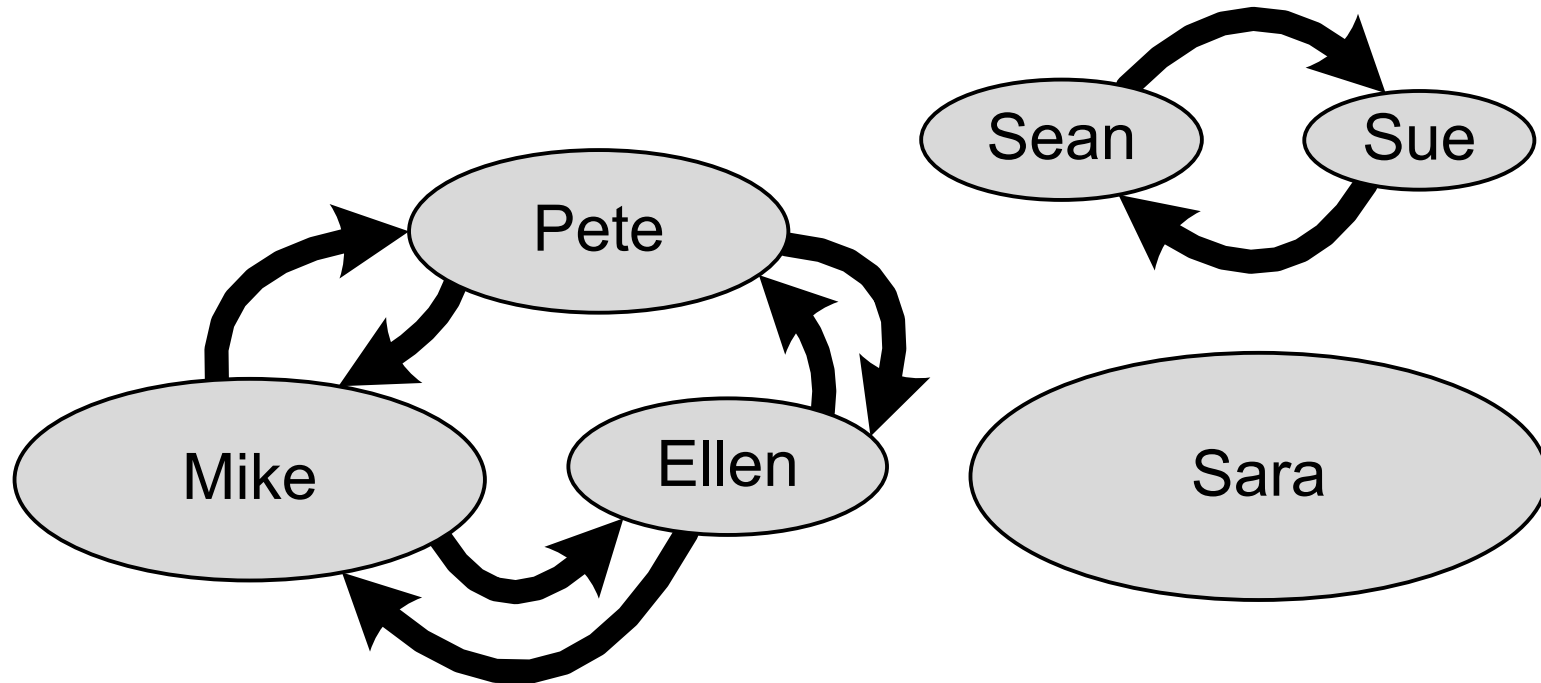
Profiles based on resource-activity matrix

	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>	<i>h</i>
Pete	0.3	0	0.345	0.69	0	0	0.135	0.165
Mike	0.5	0	0.575	1.15	0	0	0.225	0.275
Ellen	0.2	0	0.23	0.46	0	0	0.09	0.11
Sue	0	0.46	0	0	0	0	0	0
Sean	0	0.69	0	0	0	0	0	0
Sara	0	0	0	0	2.3	1.3	0	0



**find the three
profiles**

Social network based on similarity of profiles



Resources that execute similar collections of activities are related. Sara is the only resource executing e and f. Therefore, she is not connected to other resources. Self-loops are suppressed as they contain no information (self-similarity).